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The purpose of this review is not to be a 600-page immuno textbook with superfluous discussion of minutiae that will never get tested. The purpose of this doc is to focus strictly on what will increase your USMLE Step 1 score based on what shows up on NBME and Step exams. I had to try and strike a balance between sufficient background discussion of key concepts and just focusing on where your HY points are (i.e., saving you time).

- "What do I need to know about MHC molecules for USMLE?"

Simplified comparison of MHC I and MHC II for USMLE		
MEHLMANMEDICAL	MHC I	MHC II
Located on which cell surfaces?	All nucleated cells (not RBCs)	Antigen-presenting cells (APCs) – i.e., dendritic cells, macrophages, B cells
Interact with which cell?	CD8+ T cell	CD4+ T cell
Bind to what?	T cell receptor (TCR) + CD8	TCR + CD4
How is antigen loaded onto on the MHC molecule?	Associated with β2-microglobulin and TAP protein	Following release of invariant chain within acidified endosome
Purpose?	Displays intracellular pathogen to T cells (viruses, some bacteria)	Displays phagocytosed extracellular pathogen to T cells (most bacteria, fungi, parasites)
Associated drugs	**Bortezomib** is a proteasome inhibitor that ↓ ability of MHC I to present antigen to CD8+ T cells (on NBME).	Gold salts used as antiquated Tx for rheumatoid arthritis inhibit both MHC I and II

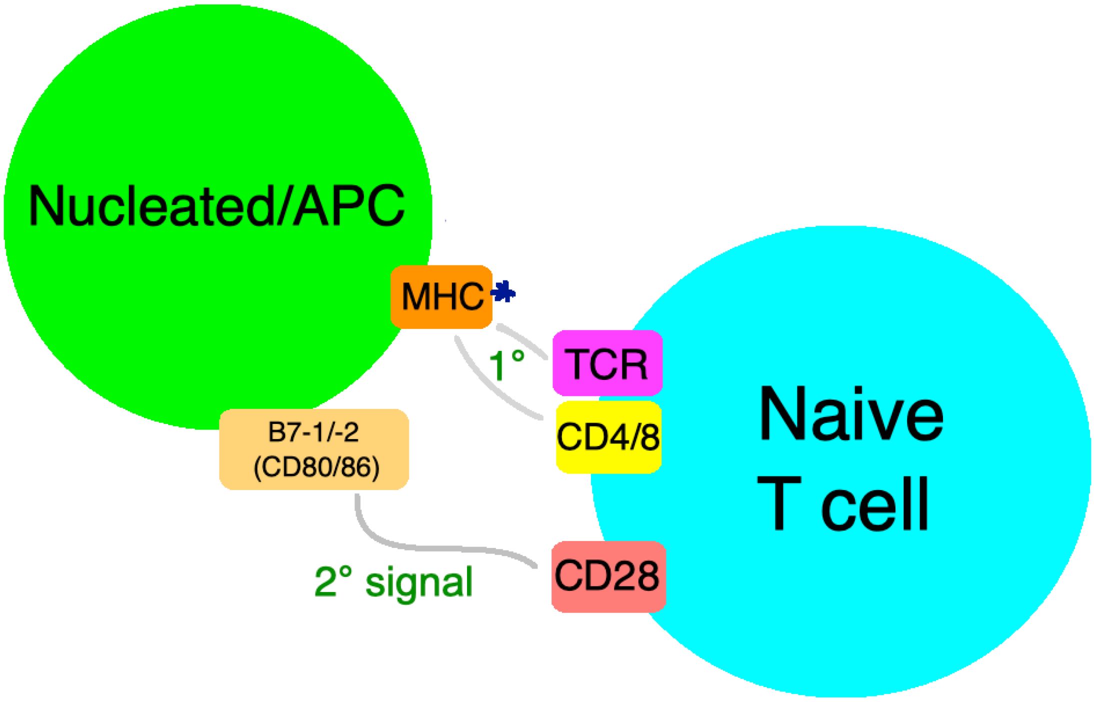
- What do I need to know about innate vs adaptive immunity for USMLE?"

Innate vs Adaptive immunity HY points		
MEHLMANMEDICAL	Innate	Adaptive
Cells + molecules involved?	Neutrophils, macrophages, NK cells, complement proteins	B cells, T cells, antibodies
Mechanism?	Germline-encoded	Created via V(D)J recombination
Onset?	Occurs rapidly (minutes to hours)	Occurs slowly (days)
Duration?	Short-lived; no memory cells produced	Longer-lived; memory cells produced

- "What do I need to know about natural killer (NK) cells?"
    - Highest yield point for USMLE is: They kill viral-infected and tumor cells in response to identifying decreased MHC-I expression. In other words, if a viral-infected or tumor cell downregulates MHC-I to try to evade the immune system, NK cells can "see" that and attack/kill the cell. Other info:
    - Part of innate immunity; stimulated by IL-12.

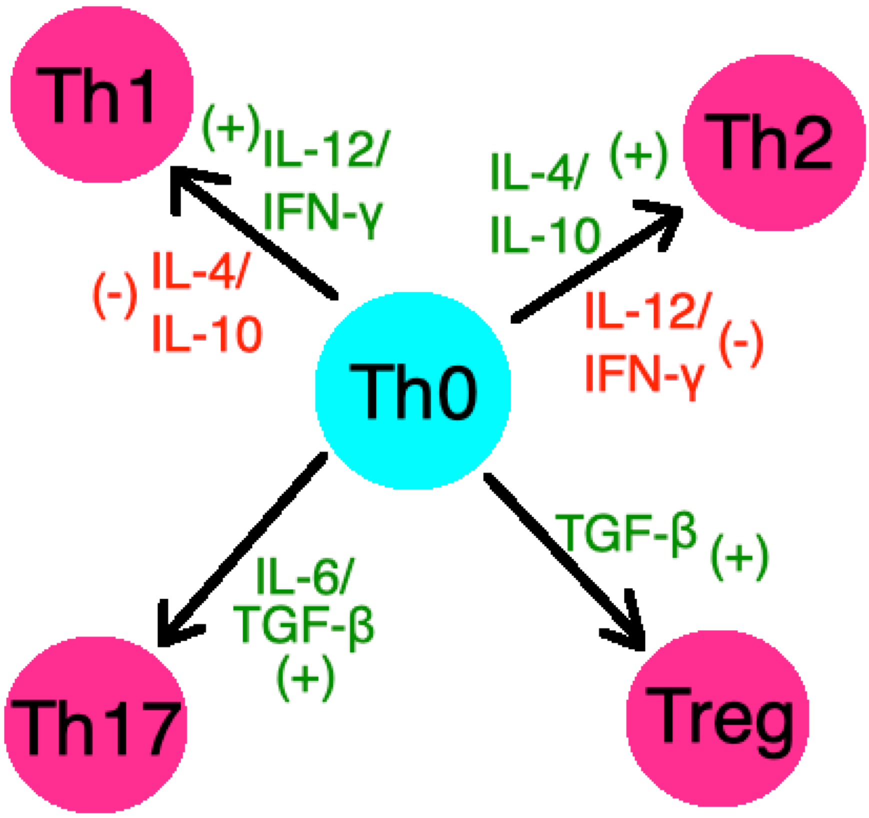
  - NK cells, B cells, and T cells are the three types of lymphocytes (all derived from lymphoid progenitors).
  - CD16 on NK cells enable a process called antibody-dependent cell-mediated cytotoxicity:
    - Antibodies produced by plasma B cells bind to antigen at their Fab regions, leaving Fc regions of the antibody facing outward.
    - CD16 on NK cells binds Fc regions of antibody.
    - Crosslinking of CD16 molecules occurs, causing the NK cell to release of granzymes, granulysins, and perforins, which induces apoptosis in target cell.
- "What do I need to know about B cells?"
  - Highest yield point: Main cell of humoral immunity. It differentiates into plasma cells, which secrete tons of antibodies.
  - Main cell of humoral immunity; differentiates into antibody-secreting plasma cell.
  - Undergoes somatic hypermutation and affinity maturation, enabling increased antibody specificity for antigen.
  - Areas within lymph nodes and the spleen where B cells proliferate and differentiate are known as germinal centers.
  - Type of antigen-presenting cell (APC), meaning it can phagocytose extracellular pathogen and present it to CD4+ T cells on MHC II.
  - Most lymphomas and leukemias are B cell in origin.
- "What do I need to know about T cells?"
  - Two main types to know for USMLE: CD8+ and CD4+.
  - Cytotoxic CD8+ T cells kill intracellular organisms (i.e., mostly viruses, *some* bacteria). Its T cell receptor (TCR) binds to MHC I on all nucleated cells. Once activated, it will release cytotoxic granzymes, granulysins, and perforins, causing apoptosis of the target (infected) cell.
  - CD4+ T cells have many types. Th0 is an upstream CD4+ T cell subtype. This can differentiate into, most importantly, Th1 and Th2. Th1 primarily activates macrophages. Th2 primarily stimulates B cells, enabling antibody production.
- "What do I need to know about T cell activation?"

- When a cell loaded with antigen on MHC I or II comes into contact with a CD8+ T cell (aka cytotoxic T cell) or CD4+ T cell (aka Helper T cell), respectively, the primary signal is merely the binding of MHC to both TCR and the CD molecule. In other words:
  - MHC I binds to TCR + CD8.
  - MHC II binds to TCR + CD4.
- A secondary signal then ensues: B7-1 (CD80) or B7-2 (CD86) on the nucleated cell / APC binds to CD28 on the naive T cell. The T cell is now activated. In other words, the second step is:
  - B7-1/-2 (CD80/86) binds to CD28.
  - This second step is the same for both CD4+ and CD8+ T cells.

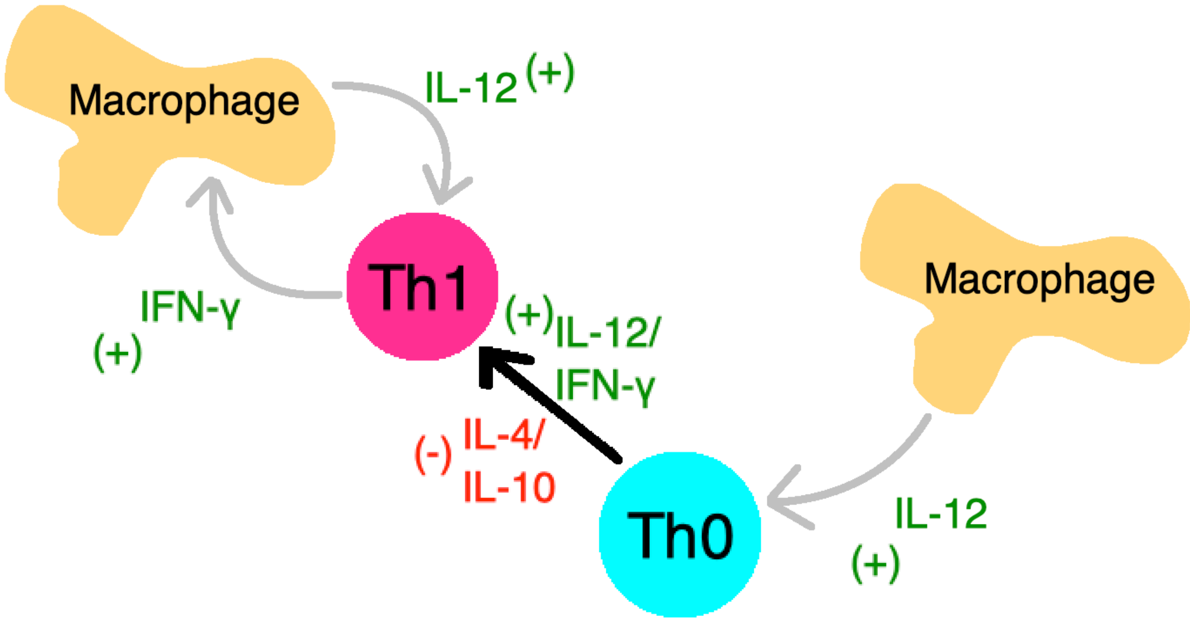


- Activation of the T cell results in:
  - CD8+ cytotoxic cells directly kill nucleated cells that have antigen loaded on MHC I.
  - CD4+ T cells are able to exert a variety of effects, depending on their differentiation (discussed below). The activation of the CD4+ T cell usually occurs after its differentiation.
- So the quick summary of T cell activation:
  - Primary signal = MHC binds to TCR/CD.
  - Secondary signal = B7 protein binds to CD28.

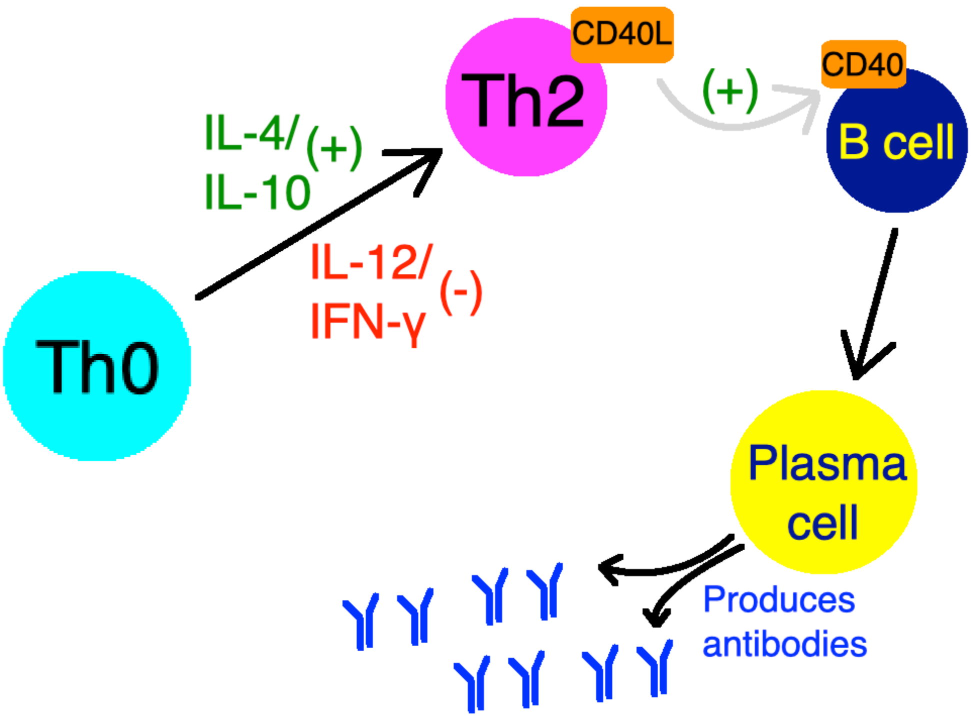
- "What do I need to know about T cell differentiation?"
  - For USMLE purposes, this applies to just CD4+ T cells. In other words, they want you to know mostly about how Th0 can become Th1 and Th2 cells. As mentioned earlier, a Th0 cell is an upstream type of CD4+ T cell that has not yet differentiated. CD8+ T cells are more simplistic and are merely activated to kill target cells, but CD4+ T cells undergo various routes of differentiation in addition to their activation.
  - Th0 can become Th1, Th2, Th17, and Treg (regulatory) cells.
  - Th1 cells stimulate macrophages, which kill phagocytosed pathogens and (if necessary) organize into granulomas.
  - Th2 cells stimulate B cells to become plasma cells, which make antibodies that kill extracellular pathogens.
  - Th17 cells stimulate neutrophilic production and recruitment.
  - Treg cells prevent autoimmunity (i.e., are a suppressive type of T cell).



- Th0 → Th1 occurs via stimulation with IL-12 and IFN-λ. IL-12 is secreted by macrophages. IFN-γ is secreted by Th1 cells. Once a Th0 cell becomes a Th1 cell, it activates macrophages via IFN-γ. Macrophages then secrete even more IL-12, which continues to stimulate more Th0 → Th1 cells. This IL-12-IFN-γ loop is HY for Step 1.



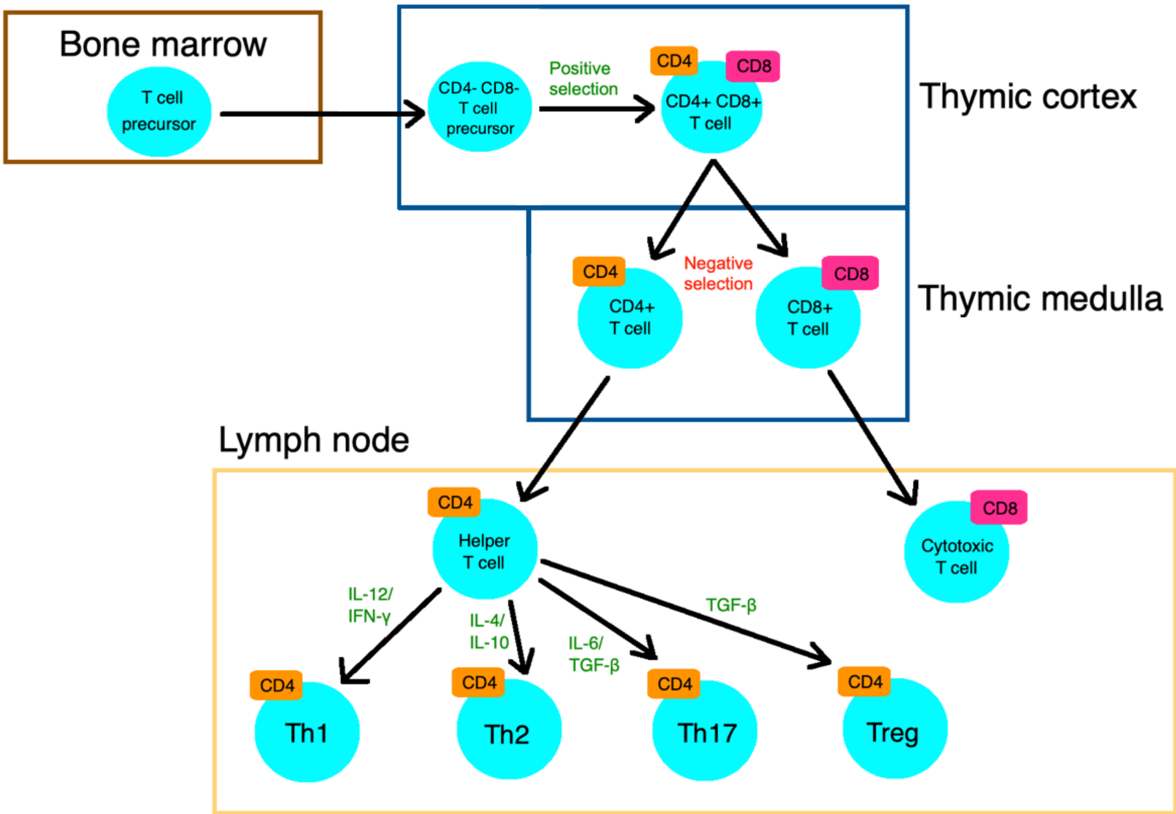
- Stimulated macrophages may or may not go on to form granulomas, which are dense clusters of organized macrophages that appear if inflammation is chronic. Activated macrophages composing a granuloma are called histiocytes, or epithelioid macrophages.
  - In short: IL-12/IFN-γ → ↑ Th1 lineage → ↑ macrophage activation.
  - Macrophages that organize into granulomas are called histiocytes, or epithelioid macrophages.
- Th0 → Th2 occurs via stimulation with IL-4 and IL-10. The Th2 cell has a ligand on its cell surface called CD40 ligand (CD40L). This binds to CD40 on the B cell, activating it and facilitating its differentiation into a plasma cell. The plasma cell (or plasma B cell) then makes copious antibodies that neutralize extracellular pathogen (humoral immunity).
  - In short: IL-4/IL-10 → ↑ Th2 = B cell activation → ↑ plasma cells → ↑ Abs.



- It should be noted that Th1 cells *also* co-stimulate macrophages via CD40L-CD40 interaction, however on USMLE, this CD40L-CD40 interaction is specifically HY with respect to Th2 activation of B cells.
- The plasma B cell is able to dramatically increase the specificity of the antibodies it produces via processes called somatic hypermutation and affinity maturation.
  - **Somatic hypermutation** refers to mutation rates in the genes of immunoglobulins that are 1,000,000 times greater than in other cell lines.
  - This results in **affinity maturation**, which means antigen-binding regions (Fab) of immunoglobulins (antibodies) are capable of generating increasingly greater affinity for pathogens.
  - Do not confuse somatic hypermutation and affinity maturation (B cell processes) with **V(D)J recombination**, which refers to B *and* T cells early in their maturation process generating antibodies and TCR, respectively, that bear a diverse array of combinations, specificities, and affinities.
- The Th1 and Th2 differentiation pathways are considered to be in opposition. When one activates, the other is suppressed. IL-12/IFN-γ suppress the Th2 lineage; IL-4 and IL-10 suppress the Th1 lineage. In other words:

- IL-12/IFN-γ → ↑ Th1 lineage + ↓ Th2 lineage.
- IL-4/IL-10 → ↑ Th2 lineage + ↓ Th1 lineage.

- "Can you explain T cell migration through the thymus? What do I need to know?"



- T cells migrate into the thymic cortex first as CD4 *and* CD8 **negative.** They then differentiate, still in the thymic cortex, into T cells that are **both** CD4+ and CD8+ (i.e., each T cell has **both** CD4 and 8 on its surface). **Positive selection** occurs in the cortex, where only T cells capable of binding self-MHC antigen survive (i.e., they're capable of generating an immune response).
- As T cells migrate from the cortex into the medulla, they lose either a CD4 or CD8, so the thymic medulla has CD4+ and CD8+ T cells (i.e., they don't simultaneously express both antigens anymore). **Negative selection** occurs in the medulla, where T cells capable of binding self-MHC antigen undergo apoptosis. This is a check to prevent autoimmunity.
- The CD8+ T cell then leaves the thymic medulla for the lymph node, where it becomes a cytotoxic T cell. It will bind to MHC I on all nucleated cells via its T cell receptor (TCR).

- The CD4+ T cell leaves the thymic medulla for the lymph node, where it becomes a Helper T cell (Th cell; [Th0 initially]), which then can become many different subtypes of CD4+ T cells, including Th1, Th2, Th17, and Treg (regulatory T cells). The TCRs on these cells bind to MHC II on antigen-presenting cells (APC).

- "What is the difference between cell-mediated and humoral immunity?"

MEHLMANMEDICAL.COM	Cell-mediated immunity		Humoral immunity
HY pathogens	Most viruses; ***Listeria***	TB, *M. leprae*, *Leishmania*, *Pneumocystis*, *Candida*	Most bacteria, Polio virus
Pathogen location	Cytoplasm	Vesicles of macrophages	Extracellular fluid (ECF)
Main cells involved	CD8+ T cell	Th1 CD4+ T cell, macrophages	Th2 CD4+ T cell, B cell
Effector T cell role	Induces apoptosis of infected cell	Activates macrophages	Activates B cells
MHC association	MHC I on infected nucleated cell	MHC II on macrophage or dendritic cell	MHC II on B cell or dendritic cell
HY steps	1) MHC I binds to TCR on CD8+ T cell; CD8 + CD3 on T cell also bind to MHC I 2) B7-1/7-2 (CD80/86) on nucleated cell binds to CD28 on CD8+ T cell 3) Activated CD8+ T cell induces apoptosis in infected cell by releasing granzymes, granulysins, and perforins	1) MHC II binds to TCR on CD4+ Th1 (or Th0) cell; CD4 + CD3 on T cell also bind to MHC II 2) B7-1/7-2 (CD80/86) on macrophage or dendritic cell binds to CD28 on CD4+ Th1 (or Th0) cell 3a) Th1 cell secretes IFN-γ, which activates macrophages to secrete IL-12, which further activates Th1 to secrete IFN-γ + stimulates more Th0 → Th1 3b) IL-12 and IFN-γ inhibit Th2 response 4) Th1 cell CD40L binds to CD40 on macrophages causing further activation + IL-12 secretion **(but on USMLE, this CD40L-CD40 interaction is most important for B cell activation by Th2 cells)**	1) MHC II binds to TCR on CD4+ Th2 (or Th0) cell; CD4 + CD3 on T cell also bind to MHC II 2) B7-1/7-2 (CD80/86) on B cell or dendritic cell binds to CD28 on CD4+ Th2 (or Th0) cell 3a) Th2 cell secretes IL-4 and IL-10, which activate more Th0 to Th2 3b) IL-4 and IL-10 inhibit Th1 response 4) Th2 cell CD40L binds to CD40 on B cells, facilitating differentiation to plasma cells for Ab production and isotype class-switching

HY USMLE disease associations	Choose cell-mediated immunity for viruses and *Listeria*. For chronic mucocutaneous candidiasis, choose "deficiency of cell-mediated immunity"	Choose "deficiency of humoral immunity" for IgA deficiency and Bruton XLA

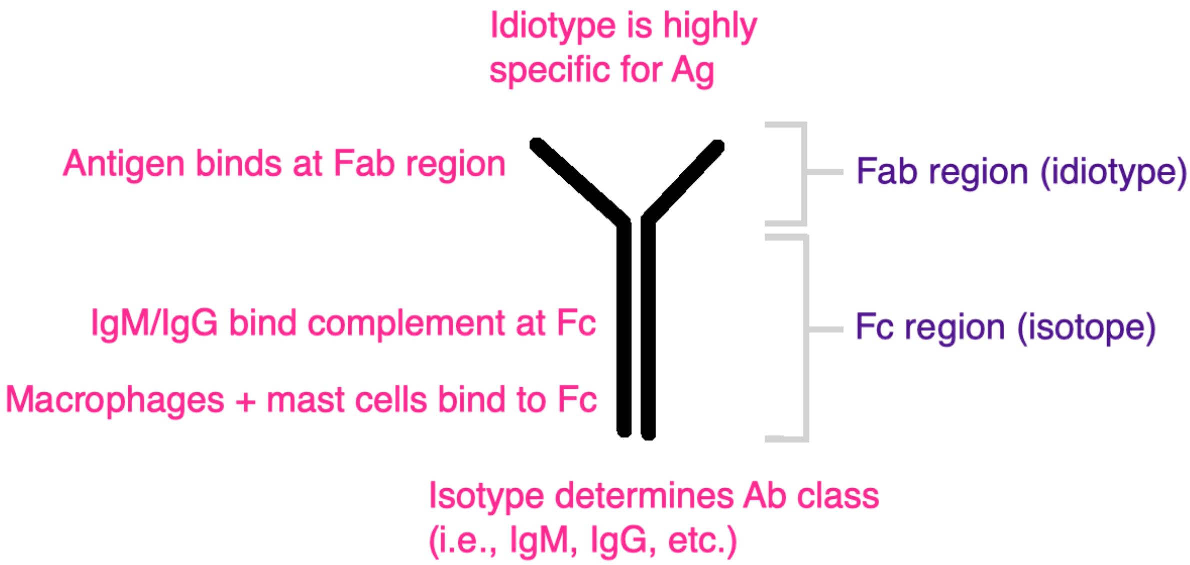
- Humoral immunity means **antibodies** are involved in the killing of the organism. If antibodies are not involved, then the process does not relate to humoral immunity.
    - Plasma B cells produce antibody and hence are the main cell of humoral immunity.
    - B cells are one type of APC that presents antigen to Th2 CD4+ T cells via MHC II.
    - Th2 cells activate B cells to become plasma cells + induce B cell isotype class-switching via CD40L-CD40 interaction (CD40L on the Th2 cell; CD40 on the B cell).
    - Therefore B cells and Th2 CD4+ T cells are integral cells in humoral immunity.
    - USMLE wants you to know that X-linked agammaglobulinemia (XLA) and IgA deficiency are deficiencies of humoral immunity – i.e., the vignette sounds like XLA, but rather than "XLA" or "IgA deficiency" as the answer, you will simply select "deficiency of humoral immunity."
- Cell-mediated immunity means "T cell-mediated" immunity that does **not** involve antibodies in order to kill the organism.
    - Cytotoxic CD8+ T cells kill nucleated cells that present intracellular antigen on MHC I molecules, where the CD8+ T cell releases granzymes, granulysins, and perforins, inducing apoptosis of the target cell. This does not involve antibody and is therefore cell-mediated, not humoral, immunity.
    - Th1 CD4+ cells activate macrophages via the IFN-γ-IL-12 loop to kill phagocytosed organisms. No antibody is made in relation to this process.
    - CD8+, Th1 CD4+ T cells, and macrophages are therefore part of cell-mediated immunity.
    - Dendritic cells are another type of APC that are part of both humoral and cell-mediated immunity, depending on if the MHC II-TCR interaction results in a Th1 (cell-mediated) or Th2 (humoral) response.
- In short:
    - Humoral immunity = B cells and Th2 CD4+ T cells.

  - Cell-mediated immunity = CD8+, Th1 CD4+ T cells, and macrophages.
  - It is inaccurate to say the difference between cell-mediated and humoral immunity is strictly CD4+ vs CD8+. This is because CD4+ Th1 cells contribute to cell-mediated immunity; CD4+ Th2 contribute to humoral immunity.

- "What do I need to know about the different antibody types and their structure?"

Antibody types + HY points				
Ab type	Binds complement?	Crosses placenta?	In breastmilk?	Function / HY points
IgM	Yes	No	No	- Primary immune responses (i.e., first Ab to appear); therefore indicates acute infection. If Q asks what would best reflect a Dx of acute congenital Toxo in a neonate, for instance, choose IgM titers in the neonate. - Pentameric in structure, therefore greatest *avidity* (i.e., has the most binding sites – 10 total [2 per Fab]). - Hyper-IgM syndrome → caused by deficiency of CD40L on Th2 CD4+ T cells, leading to ↓ binding of CD40 on B cells → differentiation into plasma B cells + ↓ isotype class-switching (i.e., plasma B cell is "stuck" at IgM and cannot make other types of Ig).
IgG	Yes	Yes	No	- Secondary immune responses (i.e., rises the most in re-infections) and chronic infections. - Has greatest *affinity* (i.e., binds most strongly to Ag); used in passive immunity (e.g., RhoGAM, HepB/VZV IVIG). - Ability for IgG to cross placenta is responsible for hemolytic disease of the newborn. - Most abundant Ab in serum (~80%).
IgA	No	No	Yes	- Most common Ab found in mucosal secretions, respiratory tract, and GIT. - Secreted as a dimer adjoined together by a J-chain (very pedantic detail that for some magical reason shows up on USMLE). - Gut-associated lymphoid tissue (GALT) consists of Peyer's patches, or aggregates of lymphoid tissue in the gut responsible for protection against intra-luminal pathogens; Peyer's patches are the predominant source of IgA-producing cells. - IgA deficiency is one of the highest yield Qs for step; most common hereditary immunodeficiency; recurrent sinopulmonary infections, atopy (i.e., seasonal allergies, asthma, eczema), ↑ risk of Giardia; associated with other autoimmune disorders like vitiligo; anaphylaxis with blood transfusion is HY detail but often omitted from NBME Qs because it's too easy. - For IgA deficiency Qs, the answer will often just be "deficiency of humoral immunity," or "deficiency of mucosal immunoglobulin." - IgA deficiency will yield false-negative Celiac screening (because one of the antibodies we normally look for is anti-tissue transglutaminase IgA); patients with Celiac have 15x greater simultaneous prevalence of IgA

				deficiency (as per above, ↑ association with other autoimmune disorders).
IgE	No	No	No	- Located on surface of mast cells; crosslinks in type I hypersensitivity responses and leads to mast cell degranulation, with release of histamine and tryptase. - Associated with allergies and eosinophilia. - Increased in helminth infections (activate eosinophils, which kill helminths by secreting major basic protein). - Hyper-IgE (Job) syndrome → **FATED** → abnormal **F**acies, *Staphylococcal* "cold" **A**bscesses, retained deciduous **T**eeth, hyper-Ig**E**, **D**ermatologic findings (i.e., eczematoid skin lesions).
IgD	No	No	No	- Part of B cell receptor. - Considered to be "obscure" and the "least important" Ab for USMLE.



- "Can you quickly explain passive vs active immunity?"

Passive vs active immunity		
MEHLMANMEDICAL	**Passive**	**Active**
Mechanism of acquisition	Receiving pre-formed antibodies	Real-life exposure to antigen
Onset	Fast (minutes to hours)	Slow (hours to days)
Duration	Short; dependent on half-life of pre-formed Abs (weeks)	Long (memory cells)
HY associations for USMLE	**Post-exposure IVIG** given for HepB, VZV, tetanus, botulism, diphtheria, rabies; IgA in breastmilk; transplacental IgG	**Vaccinations**; real-life infection

- "What do I need to know about the complement cascade/pathways? I hear about complement a lot but I'm not really sure what it does. Can you explain that."

- No, you do not need to memorize / draw out the complement pathways for Step 1. Waste of time. You just need to have a broad sense of the role complement plays in the innate immune system + some factoids about the different complement molecules.
- Complement proteins are acute phase proteins produced by the liver. There are three pathways for complement activation; they all lead to target cell death via formation of the membrane attack complex (MAC), composed of C5b-C9 complement proteins, or via stimulating opsonization and phagocytosis.
- Three main pathways: classic, alternative, and lectin:
  - Classic → when complement (C1) binds to the Fc region of IgM and IgG, it flags the antigen bound to the Ab's Fab for opsonization and phagocytosis, or cell death via MAC.
  - Alternative → triggered by C3b binding directly to the microbial cell surface.
  - Lectin → triggered by mannose-binding lectin (a type of carbohydrate-binding protein instrumental in innate immunity) binding to the microbial cell surface.
- C3b and IgG are the immune system's two major opsonins (which means they bind to antigen and flag it for phagocytosis).
  - Do not confuse this with IgM and IgG's ability to bind complement. Phagocytes do not have an Fc receptor for IgM, making IgM a poor opsonin. In other words, whilst IgM and IgG bind complement, it is IgG and C3b that are the immune system's major opsonins.
- C3 goes down in post-streptococcal glomerulonephritis (PSGN) and SLE flares.
- C4 goes down in cryoglobulinemia (discussed later).
- C3a, C4a, and C5a are known as anaphylatoxins, which can trigger degranulation of mast cells, causing anaphylaxis.
- C5a is a neutrophilic chemotactic molecule (as with IL-8, LTB-4, kallikrein, platelet-activating factor, and bacterial products).
- Terminal complement deficiency (C5-9) is associated with recurrent *Neisseria* infections (both gonococcal and meningococcal).

  - C1 esterase inhibitor deficiency (not C1 esterase; C1 esterase *inhibitor*) is associated with hereditary angioedema.
  - Paroxysmal nocturnal hemoglobinuria is caused by "increased complement-mediated lysis" of RBCs. It is caused by deficiency of GPI anchor, which "anchors" CD55 (decay-acceleratory factor) and protectin (CD59) to the RBC cell surface. CD55/59 protect the RBC from complement-mediated lysis.
- "Can you explain thymus-independent vs thymus-dependent antigens?"
  - Thymus-independent means an antigen is **not** displayed on MHC molecules. Therefore there is no resultant T cell response.
  - Thymus-dependent means an antigen **is** displayed on MHC molecules. Therefore there is a resultant T cell response.
  - Antigens need to be proteinaceous in order to be displayed on MHC molecules and generate a T cell response. In other words, if a Q asks what kind of molecule could induce a T cell response, choose whatever answer is the protein, even if it sounds weird (e.g., flagellin, or poly-D-glutamic acid capsule of *C. anthracis*). Lipopolysaccharide (LPS) or the polysaccharide capsule of *H. influenzae* type B, for instance, are wrong, since they are not protein.
  - If an antigen is not proteinaceous, then it will not be expressed on MHC molecules (i.e., thymus-independent antigen). These antigens therefore require conjugation to a protein so that a T cell response can be elicited. USMLE wants you to know that *H. influenzae* type B, *Strep pneumoniae*, *N. meningitidis,* and typhoid (*Salmonella typhi*) are **conjugate vaccines** that are **polysaccharide-based** (on NBME for Step 1) – i.e., the polysaccharide capsule from these organisms is conjugated to a proteinaceous substrate. Toxoid is the wrong answer for these vaccines.
  - Toxoids are vaccines in which an inactivated toxin from an organism, which is proteinaceous, is administered. Diphtheria and tetanus are classic examples.
  - Thymus-independent response involves an antigen binding to B cell receptor (BCR), followed by the B cell making antibody against the antigen without phagocytosing + expressing it on MHC II to Th2 CD4+ cells. This process does not result in any type of memory cell being produced, so immunity is short-lived.

  - In contrast, thymus-dependent responses lead to the production of memory cells, so the immunity is longer-lived.
- "What do I need to know about the various cytokines?"
  - Memorize macrophages as secreting, most importantly: TNF-α, IL-1, IL-6, IL-8, and IL-12.
  - Memorize IL-2 as a T cell stimulator.
  - Memorize TNF-α and IL-1 as the two associated with sepsis: TNF-α causes low BP; IL-1 causes fever.
  - Memorize IL-8 as a neutrophilic chemotactic molecule.
  - Interferons can broadly facilitate with killing viral infections by inhibiting viral replication.

Cytokine HY points for USMLE	
TNF-α	- Increases vascular permeability + causes ↓ systemic vascular resistance and ↓ BP in septic shock. - If the USMLE asks which cytokine is most responsible for patient's presentation of septic shock, TNF-α is the answer. - Also known as cachexia factor → is the main cytokine responsible for weight loss in the setting of malignancy (HY). The Q can ask for TNF-alpha directly, but I've also seen the answer for the weight loss as "cytokine effect," where "paraneoplastic effect" is wrong answer.
IL-1	- Causes fever. - If the Q tells you a guy with sepsis has low BP and fever, they want TNF-α and IL-1 as the two cytokines responsible; if you're forced to choose only one cytokine for septic shock, choose TNF-α, but if they ask specifically about the fever, choose IL-1. - If the Q tells you an NSAID is used to decrease fever, choose prostaglandin, not IL-1, as the molecule responsible for fever. - Also known as osteoclast-activating factor; can contribute to osteoporosis and lytic lesions of bone in the setting of metastases and multiple myeloma; USMLE Q might show you a pic of an osteoporotic vertebra, and then the answer is just "increased IL-1 activity." IL-6 also plays a role in osteoporosis, but NBME for Step 1 has IL-1 as many answers.
IL-2	- Stimulates T cells; when you think of IL-2, the first thing that should come to mind is, "Ok, that's the one that broadly activates T cells." **- Cyclosporin** and **tacrolimus** are HY drugs that decrease *transcription* of IL-2 via inhibiting calcineurin. **- Sirolimus** decreases *responsiveness* to IL-2. - Aldesleukin is a recombinant IL-2 used in the Tx of RCC and metastatic melanoma (essentially, ↑ T cell response helps fight cancer, and IL-2 ↑ T cell response).
IL-3	- Stimulates myeloid lineage (i.e., ↑ mostly neutrophils); think of IL-3 as similar to GM-CSF.
IL-4	- Both IL-4 and IL-10 stimulate Th0 → Th2, and inhibit Th0 → Th1. - Th2 $CD4^+$ T cells normally stimulate B cells, which mature into plasma cells that secrete immunoglobulins. This means this cytokine is important for supporting humoral immunity. - Th1 normally stimulate macrophages, which do not secrete immunoglobulins and are part of cell-mediated immunity.
IL-5	- Stimulates eosinophil maturation.
IL-6	- Stimulates acute-phase protein release from the liver (i.e., CRP, hepcidin). - Can cause osteoporosis (in addition to IL-1). - Tocilizumab is a monoclonal Ab against IL-6 receptor used in severe rheumatoid arthritis. - IL-6 and TGF-β stimulate Th0 → Th17.
IL-8	- Neutrophilic chemotactic molecule - IL-8, C5a, LTB-4, platelet-activating factor, kallikrein, and bacterial products all stimulate neutrophilic chemotaxis.

IL-10	- HY as an anti-inflammatory cytokine that plays a role in suppressing various immune responses. If the USMLE asks which cytokine can decrease an immune response, choose IL-10 (or TGF-β). - Both IL-10 and IL-4 stimulate Th0 → Th2, and inhibit Th0 → Th1. - Therefore IL-10 functions to ultimately increase CD4+ activation of B cells. - This means this cytokine is important for supporting humoral immunity.
IL-11	- Stimulates platelets. - Oprelvekin is a recombinant IL-11 that can be used to stimulate thrombopoiesis post-chemotherapy.
IL-12	- Both IL-12 and IFN-γ stimulate Th0 → Th1, and inhibit Th0 → Th2. - Stimulates Th1 cells to make IFN-γ. - IL-12 receptor deficiency is associated with ↑ TB infections (i.e., ↓ IL-12 response → ↓ IFN-γ → ↓ macrophage activity → ↑ risk for TB infections); treat with IFN-γ. - Stimulates NK cells.
IFN-γ	- HY cytokine produced by Th1 CD4+ T cells that stimulates macrophages. - Macrophages stimulated by IFN-γ produce IL-12. - Both IFN-γ and IL-12 stimulate Th0 → Th1, and inhibit Th0 → Th2. - Used in the Tx of NADPH oxidase deficiency (chronic granulomatous disease). - Also given to patients with IL-12 receptor deficiency.
IFN-β	- Used *between* flares of multiple sclerosis to decrease recurrence (for acute flares gives steroids).
TGF-β	- Anti-inflammatory cytokine, similar to IL-10 (i.e., when you think of TGF-β and IL-10, think ↓ **inflammation**). - TGF-β and IL-6 stimulate Th0 → Th17. - TGF-β stimulates Th0 → Treg.

- "What should I know about various cell surface markers?"

Cell surface marker HY points for USMLE	
CD1a	- Marker in Langerhan cell histiocytosis, which presents as lytic lesions in a patient where biopsy of various organs can show Birbeck granules (tennis racquet or rod-shaped granules).
CD3	- Generic T cell marker; seen on all T cells. Shows up occasionally in immuno stems. - Associated with TCR; binds to target cell MHC + either CD4 or CD8.
CD4	- Marker for T helper cells (Th0, Th1, Th2, Th17, Treg); binds to MHC II on APCs (B cells, macrophages, dendritic cells).
CD5 + CD23	- Seen in CLL (sounds LY and pedantic but I've seen these in CLL vignettes on NBME).
CD8	- Marker for cytotoxic T cells; binds to MHC I on nucleated cells.
CD10	- Associated with ALL.
CD14	- On macrophages; associated with toll-like receptor 4. - Binding of the lipid A of LPS (endotoxin) to CD14 results in macrophage release of TNF-α + IL-1 → septic shock caused by endotoxin. - Do not confuse with septic shock caused by *S. aureus* TSST superantigen bridging TCR and MHC II → toxic shock syndrome.
CD15 + CD30	- Seen on Reed-Sternberg B cells in Hodgkin disease.
CD16 + CD56	- Both on NK cells. - CD16 binds Fc region of IgG and causes antibody-dependent cell-mediated cytotoxicity → lysis of target cell by NK cell. - CD56 is just a specific NK cell marker.
CD18	- CD18 is a subunit of LFA-1 integrin found on various WBCs. - Deficiency of CD18/LFA-1 integrin causes leukocyte adhesion deficiency.
CD19 + CD20	- Specific B cell markers; **rituximab** is a monoclonal Ab against CD20. - Q can mention a virus invading CD19/20 (+) cells → answer is EBV (invades B cells).
CD21	- Receptor for EBV entrance into B cells.
CD25	- Regulatory T cell (Treg) marker
CD28	- On T cells; bound by B7-1/7-2 (CD80/86) on nucleated cells/APCs as secondary signal following MHC primary interaction.

CD34	Ubiquitous on hematopoietic stem cells (i.e., not specific marker for, but found on all).
CD40	- On surface of B cells. - CD40-ligand (CD40L) on CD4+ T cells binds to CD40 on B cells. - CD40L-CD40 interaction between the CD4+ T cell and the B cell is responsible for isotype class-switching + B cell differentiation into plasma cells. - Deficiency of CD40L causes hyper-IgM syndrome.
CD55 + CD59	- Deficient in paroxysmal nocturnal hemoglobinuria.
CD80 + CD86	- Also known as B7-1/7-2. - Found on nucleated cells/APCs. - Bind to CD28 on T cells as secondary signal following primary MHC interaction.

- "What autoantibodies do I need to know for USMLE?"

HY autoantibodies for USMLE	
HY disease association	**HY antibodies**
Antiphospholipid syndrome	- Anti-β2-microglobulin; anti-cardiolipin; lupus anticoagulant. - The latter is the name for either of the former two if the patient happens to have SLE. - Presents as *in vivo* thromboses despite an ↑ *in vitro* aPTT. - These antibodies are associated with recurrent miscarriage. - Can cause false-positive syphilis screening (i.e., SLE patient who gets positive syphilis VDRL test).
Bullous pemphigoid	- Anti-hemidesmosome (bullous pemphigoid antigen). - Hemidesmosomes connect the dermis to epidermis at the basement membrane. - In BP we see linear immunofluorescence on skin biopsy because hemidesmosomes line the basement membrane. - Q can say there are sub-epidermal blisters (pemphigus vulgaris, in contrast, the Q can say intra-epidermal blisters). - Usually no oral involvement, but not mandatory rule. - No Nikolsky sign (i.e., no sloughing of skin with friction).
Pemphigus vulgaris	- Anti-desmosome (anti-desmoglein 1 and 3). - Desmosomes connect adjacent epidermal cells side-to-side. - In PV we see a net-like immunofluorescence on skin biopsy because the desmosomes are between / outline the skin cells. - Q can say there are intra-epidermal blisters (bullous pemphigoid, in contrast, the Q can say sub-epidermal blisters). - Can present with oral involvement, albeit not mandatory. - (+) Nikolsky sign (i.e., sloughing of skin with friction).
Goodpasture syndrome	- Anti-collagen IV (anti-GBM; glomerular basement membrane). - Goodpasture presents as male 20s-40s with hematuria + hemoptysis + anti-collagen IV antibodies.

	- Causes linear immunofluorescence on renal biopsy, since collagen IV lines basement membranes. - Don't confuse with Alport syndrome, which is XR disorder due to *mutations* in type IV collagen (not Ab). - Alport presents as male with red urine + eye/ear problem.
Granulomatosis with polyangiitis (Wegener)	- c-ANCA, aka anti-proteinase 3 (anti-PR3). - Dumb mnemonic I created that helps some of my students: **W**ater **C**loset (**W**egener **C**-ANCA). - Presents are hematuria + hemoptysis + "head-itis" (i.e., problem with the head, e.g., nasal septal perforation, mastoiditis, sinusitis, necrotizing otitis). - Sometimes has neuro findings (mononeuritis multiplex).
Eosinophilic granulomatosis with polyangiitis (Churg-Strauss)	- p-ANCA, aka anti-myeloperoxidase (anti-MPO). - Presents as asthma + eosinophilia. - Sometimes has neuro findings (mononeuritis multiplex).
Microscopic polyangiitis	- p-ANCA (anti-myeloperoxidase). - Presents as usually just red urine in person with (+) p-ANCA. - Sometimes has neuro findings (mononeuritis multiplex).
Systemic lupus erythematosus (SLE)	- Anti-double-stranded DNA (dsDNA). - Anti-Smith (ribonucleoprotein). - Anti-hematologic cell line Abs. - Lupus anticoagulant. - Should be noted that anti-dsDNA goes up in acute flares + best reflects renal prognosis. - Anti-Smith is *more specific* than anti-dsDNA, meaning it shows up less often than anti-dsDNA, but if it does show up, the patient absolutely is ruled-in for having SLE. - Thrombocytopenia is an exceedingly HY finding in SLE due to the anti-hematologic cell line Abs. - If all cell lines are down in SLE, aplastic anemia (i.e., ↓ bone marrow production) is wrong answer; choose "increased peripheral destruction" as answer instead, since the cause is merely antibodies against the cells. - Lupus anticoagulant I discussed above in anti-phospholipid syndrome; causes recurrent miscarriages and false (+) syphilis tests. - Most common presenting feature in SLE overall is arthritis. The Q need not mention malar rash (I'd say only ~1/3 of Qs). So if you get, e.g., 34-year-old woman with arthritis and thrombocytopenia + no other info, that is pass-level SLE till proven otherwise. - Complement protein C3 can be down in SLE flares. - Renal diagnosis for lupus nephritis is basically always diffuse proliferative glomerulonephritis (DPGN) on NBME.
Drug-induced lupus (DIL)	- Anti-histone antibodies. - Caused by various drugs (**M**om is **HIPP**): **M**inocycline, **H**ydralazine, **I**NH, **P**rocainamide, **P**enicillamine. - Presents usually as arthritis + pericarditis or mediastinitis in patient who was given one of the above buzzy meds.

Myasthenia gravis	- Post-synaptic nicotinic acetylcholine receptor antibodies. - Sometimes a paraneoplastic of thymoma, but need not be. - Presents usually as 40s woman with triad of diplopia, eyelid ptosis, and dysphagia or hoarseness of voice. Gets worse with activity, whereas Lambert-Eaton improves. - USMLE wants pyridostigmine (cholinesterase inhibitor) as Tx.
Lambert-Eaton	- Presynaptic voltage-gated calcium channel antibodies. - Sometimes a paraneoplastic of small cell lung cancer. - Presents as proximal muscle weakness that improves with activity (e.g., patient can stand up from chair after trying a few times).
Small cell cerebellar dysfunction	- Anti-Hu/-Yo. - Presents as ataxia in someone with small cell lung cancer and negative CNS imaging (meaning metastases to brain aren't the cause of the neurologic issues).
Polymyositis / Dermatomyositis	- Anti-Jo1. - Polymyositis and dermatomyositis are usually idiopathic autoimmune conditions, but they can sometimes be a paraneoplastic of ovarian cancer. - Present as proximal muscle weakness on physical examination + increased serum CK. - Dermatomyositis = polymyositis + skin findings. The latter are shawl rash, Gottron papules, mechanics hands, and heliotrope rash.
Limited-type systemic sclerosis / scleroderma	- Anti-centromere. - Scleroderma = another name for systemic sclerosis. - Limited type merely = CREST syndrome = Calcinosis, Raynaud phenomenon, Esophageal dysmotility, Sclerodactyly, Telangiectasias. - Pulmonary fibrosis leading to pulmonary HTN is HY for both limited and diffuse types of scleroderma.
Diffuse-type systemic sclerosis / scleroderma	- Anti-topoisomerase I (Scl-70). - Can present as same findings as in limited-type, but also with malignant HTN (e.g., BP of 220/120) due to renal involvement causing a surge of RAAS.
Primary biliary cirrhosis	- Anti-mitochondrial. - Presents as woman 20s-50s with ↑ ALP, direct bilirubin, and serum cholesterol, with generalized pruritis and Hx of other autoimmune disease in her or family member (e.g., RA, SLE).
Autoimmune hepatitis	- Anti-smooth muscle. - Presents as woman 20s-30s with idiopathic elevation of hepatic transaminases. - Diagnosis of exclusion (meaning you have to eliminate the other answer choices to get there).
Rheumatoid arthritis	- Rheumatoid factor (an IgM Ab against the Fc region of IgG). - Anti-cyclic citrullinated peptide (CCP) antibodies. - Anti-CCP is *more specific* than rheumatoid factor for RA.
Sjögren syndrome	- Anti-SS-A (anti-Ro); anti-SS-B (anti-La). - Presents as xerostomia (dry mouth) + xerophthalmia (dry eyes) + arthritis.
Graves disease	- Anti-TSH receptor; this antibody is called thyroid-stimulating immunoglobulin, or TSI, and *activates* the TSH receptor. - Most common cause of hyperthyroidism. - Exophthalmos and pretibial myxedema are HY findings.
Hashimoto thyroiditis	- Anti-thyroperoxidase, aka anti-microsomal antibodies. - Anti-thyroglobulin.

	- Highest yield cause of hypothyroidism in Western countries. - Worldwide the most common cause is iodine deficiency. - Buzzy findings are weight gain, brittle hair, dry/doughy skin, and cold intolerance. The USMLE often will omit these findings because they're too easy. - HY additional findings I've seen show up in NBME Qs are: bradycardia (55-60), menstrual irregularity, dysthymia/depression/apathy, carpal tunnel syndrome (due to glycosaminoglycan [GAG] deposition), ↑ cholesterol, proximal muscle weakness with ↑ creatine kinase (hypothyroid myopathy), transaminitis (sounds weird, but LFTs can be ↑). - Lymphocytic infiltrate is seen on Hashimoto thyroid biopsy.
Pernicious anemia	- Anti-parietal cell or anti-intrinsic factor antibodies. - Presents as B12 deficiency (intrinsic factor is needed to bind to B12 so it can be absorbed at the terminal ileum).
Celiac disease	- Anti-endomysial (aka anti-gliadin). - Anti-tissue transglutaminase IgA (antibody screening will yield false-negatives if patient also has IgA deficiency). - Insensitivity to wheat, oats, rye, and barley. - Duodenal biopsy shows flattening of the intestinal villi on biopsy. Can present with iron deficiency anemia.
Type I diabetes mellitus	- Anti-glutamic acid decarboxylase (anti-GAD). - Anti-zinc transporter 8.
Primary membranous glomerulonephritis	- Anti-phospholipase A2 receptor. - Membranous glomerulonephritis is a nephrotic syndrome. - Other HY causes of MG are drugs (gold salts, dapsone, sulfonamides), autoimmune diseases like RA, cancer (e.g., pancreatic, breast), and hepatitis B. So any of these patients + nephrotic syndrome = MG usually.
Thrombotic thrombocytopenic purpura (TTP)	- Anti-ADAMTS13, which is a matrix metalloproteinase that cleaves vWF multimers. - Presents as pentad of 1) thrombocytopenia, 2) schistocytosis (the combo of 1+2 is called microangiopathic hemolytic anemia; MAHA), 3) renal issues (hematuria or ↑ RFTs), 4) fever, 5) neurologic signs.
Immune thrombocytopenic purpura (aka idiopathic thrombocytopenic purpura; ITP)	- Anti-GpIIb/IIIa on platelets (platelet aggregation). - Presents classically as school-age kid who gets petechiae and/or nose bleeds following a viral infection. - Can also present in random woman 30s-40s with ↑ bleeding time and ↓ platelet count. - The Q need not mention the viral infection, but if they do, then the Q is hyper easy and pass-level.
Heparin-induced thrombocytopenia (HIT)	- Antibodies against platelet factor 4-heparin complex (i.e., anti-PF4-heparin). - Presents as (you'd never guess) thrombocytopenia following heparin administration (e.g., for DVT or PE). - Treatment is direct thrombin inhibitor (e.g., dabigatran); warfarin is wrong answer for Tx.
Addison disease	- Can sometimes be caused by anti-21-hydroxylase Abs. - Low Na^+, high K^+, low bicarb, low pH, high ACTH. - Hyperpigmentation common.
Rheumatic heart disease	- Anti-myosin; anti-valve-derived proteins. - Should be noted that these Abs are formed against Group-A Strep (*S. pyogenes*) M protein and almost always cross-react with the mitral valve (molecular mimicry).

	- Type II hypersensitivity. Don't confuse with PSGN, which is type-III hypersensitivity. - Patient will have mitral regurg early but mitral stenosis late. - Sydenham chorea is autoimmune condition of basal ganglia. - Erythema marginatum is characteristic skin condition.
Neuromyelitis optica (Devic syndrome)	- Anti-aquaporin 4 (asked on 2CK Neuro form). - Presents basically the same as multiple sclerosis except they'll tell you these Abs are positive and that the CSF is negative for IgG oligoclonal bands (normally seen in MS).
Cold autoimmune hemolytic anemia	- IgM against RBCs (MMM ice cream; since ice cream is cold). - CMV and *Mycoplasma* are HY infectious associations. - (+) Coombs test (just means antibodies against RBCs).
Warm autoimmune hemolytic anemia	- IgG against RBCs. - Caused by various miscellaneous drugs and infections. - Associated with chronic lymphocytic leukemia (CLL). - (+) Coombs test (just means antibodies against RBCs).

Hemolytic disease of newborn types	
Hemolytic disease of the newborn (Rh type)	- Anti-rhesus factor (Rh). Mechanism: - Rh(-) mom in first pregnancy is exposed to fetal blood that is Rh(+). - Mom makes Abs against Rh. - Subsequent pregnancy results in IgG against Rh crossing placenta and targeting fetal Rh(+) RBCs, leading to hemolysis in the fetus. - Can range in severity, from fatal *in utero* with fetal hydrops, to more benign, where there is merely pathologic jaundice in neonate.
Hemolytic disease of the newborn (ABO type)	- Most anti-A and -B Abs people have to opposing blood types are IgM. People with O blood, however, can have fractionally greater IgG in comparison to IgM. - If mother with O blood has higher % of anti-A and -B Abs that are IgG, fetus can be symptomatic due to transplacental movement of IgG attacking the fetal RBCs. - USMLE will give you O+ mom usually in first pregnancy and A or B fetus. This is because it can of course occur in O-blood Rh- women in second pregnancies, but the USMLE wants to assess you specifically know the ABO type of HDN, so they'll give O+ mom in first pregnancy.

Transfusion reactions	
Febrile non-hemolytic transfusion reaction (FNHTR)	- Caused by activation of donor WBCs when they come into contact with antibodies in the recipient's plasma. - This interaction triggers the release of cytokines from donor WBCs. - These cytokines then cause symptoms of fever and chills. - Coombs test is **negative,** since the process doesn't involve antibodies against donor RBCs. A (+) Coombs test means there are Abs against RBCs. - The vignette will be someone who gets a blood transfusion, and then within 2 hours gets fever + chills. They will then ask for the mechanism as the answer. I've seen them write it one of two ways: 1) Preformed antibodies against donor leukocyte antigens; and 2) Cytokine release from transfused blood. - Leukoreduction, a process that removes or reduces the number of WBCs from blood components before transfusion, can be used to mitigate risk of

	FNHTR by reducing probability of donor WBCs coming into contact with recipient antibodies.
Hemolytic (ABO mismatch)	- Caused by recipient antibodies against A and/or B antigens on donor RBCs. - This causes RBC agglutination and intravascular hemolysis, causing the release of hemoglobin into the bloodstream. - Binding of recipient antibodies to donor RBCs can also lead to complement activation and lysis of RBCs. - LDH can be increased in setting of hemolysis (RBCs are packed with LDH). - The vignette will be someone who gets fever, chills, and often flank pain within minutes of a blood transfusion. - Vignette can say, "10 minutes after the transfusion is started, the patient feels an 'impending sense of doom' and the transfusion is stopped" (sounds like MI, but NBME writes this). They then ask for next best step, which is antiglobulin test (another way of writing Coombs test). - Coombs test is (+) because we have antibodies against RBCs.
Delayed transfusion reaction	- Caused by recipient antibodies against minor antigens on donor RBCs, such as Rh, Kell, Duffy, Kidd. - Coombs test is (+) because we have antibodies against RBCs. - Vignette will be someone who had surgery + received RBCs intraoperatively + over the course of a week following the surgery, hemoglobin gradually decreases. The answer is then "Coombs (+); unconjugated bilirubin" as the combo that would be seen.
Transfusion-associated lung injury (TRALI)	- The answer on USMLE if a patient has an ARDS-like presentation with bilateral crackles and low O2 sats **<6 hours** following a transfusion. - Mechanism is abnormal priming of neutrophils in the lung that react to cytokines within transfused blood products (this is straight from a CMS form explanation). - Technically a type of non-cardiogenic pulmonary edema.

Transfusion-associated circulatory overload (TACO)	- Aka transfusion-induced hypervolemia.
	- Differs from TRALI in that this is a type of cardiogenic pulmonary edema (i.e., the left heart can't handle the ↑ hydrostatic pressure from ↑ volume, so transudation into the alveoli occurs).
	- Annoying diagnosis but presents two ways:
	1) If a patient *with Hx of heart failure or MI* develops respiratory distress following transfusion of repeated blood products.
	    - A 2CK NBME Q gives a 72-yr-old with Hx of MI ten years ago who gets shortness of breath and bilateral crackles 30 minutes after transfusion with crystalloid solution and 4 packs of RBCs (they don't specify the volume of crystalloid in the Q).
	2) They don't mention Hx of cardiovascular disease + the vignette will sound exactly like TRALI **> 6 hours** following a transfusion.
	    - Surgery form for 2CK gives an elderly dude who received only 3 packs of RBCs + he develops bilateral crackles and low O2 sats, but they say this occurs 12 hours after admission (not <6 hours as with TRALI), where the answer is "X-ray of the chest" as the next best step in diagnosis. The Q doesn't rely on you discerning TACO vs TRALI to get it right, but the explanation says it's TACO.
	
	In summary, if ARDS-like picture after transfusion:
	
	If <6 hours → TRALI.
	If >6 hours → TACO.
	If heart disease → TACO regardless of time frame.

Hypersensitivity types for USMLE	
Type	**Mechanism**
I	- Immediate (within minutes): Antigen binds to Fab region of IgE on the surface of mast cell and basophils. This causes adjacent IgE crosslink on the cell surface to crosslink. The mast cell / basophil then degranulates and secretes histamine + tryptase. - Late (within hours): Cytokines recruit inflammatory cells (e.g., eosinophils) → more inflammation. - HY examples are anaphylaxis (i.e., bee sting, peanut allergy), atopy / asthma.
II	- Antibody production against one's own cells, tissues, receptors. - HY examples are Graves disease, Hashimoto thyroiditis, Goodpasture syndrome, pernicious anemia, HIT, ITP, rheumatic fever, myasthenia gravis, Lambert-Eaton, bullous pemphigoid, pemphigus vulgaris.
III	- Antibody-antigen complexes (i.e., antibodies simply bind to antigen and deposit; the antibodies do not target cells, tissues, or receptors). - HY examples are post-streptococcal glomerulonephritis, polyarteritis nodosa, SLE, arthus reaction, serum sickness.
IV	- T cell response (only HS type not associated with antibodies); can be mediated by either CD8+ or CD4+ T cells. - HY examples are PPD skin test for TB, contact dermatitis (poison ivy/oak/sumac, sunscreen, nickel on watches, latex), graft-vs-host disease.

- "What about vaccine types? Are those important?" Yeah.
    - Killed vaccines:
        - **RIPP A**lways → **R**abies, **I**nfluenza (IM), **P**olio (Salk; IM), **P**ertussis, Hep **A**.
        - **A TRIPP** could **Kill** you → Hep **A**, **T**yphoid (IM), **R**abies, **I**nfluenza (IM), **P**olio (Salk; IM), **P**ertussis, **Kill**ed vaccines.
        - Generate a humoral (antibody) response via expression on MHC II following phagocytosis by APCs, since cannot directly invade nucleated cells for expression on MHC I.
    - Live-attenuated:
        - Yellow fever, varicella, rotavirus (oral), influenza (intranasal), Polio (Sabin; oral), MMR, Typhoid (oral).
        - Will be expressed on MHC I by nucleated cells → generates CD8+ T cell-mediated response.
        - MMR and varicella can be given to patients with HIV if CD4 count >200 cells/uL.
        - Avoid live-attenuated vaccines in pregnancy.
    - Recombinant: HPV, HBV.
    - Toxoid: Diphtheria, tetanus.
- "What are the highest yield points about the immunodeficiencies?"

Immunodeficiencies HY points	
Severe-combined immunodeficiency (SCID)	- X-linked recessive most common → usually due to common gamma chain mutation, or mutation in IL-2 receptor. - Autosomal recessive type less common → due to adenosine deaminase (ADA) deficiency. - Combined T and B cell deficiency. - T cell deficiency → absent thymic shadow. - B cell deficiency → scanty/absent lymph nodes and tonsils. - Q will give you sick kid 1-5 yrs-old who's had infections of all different types (i.e., viral, fungal, bacterial, protozoal) **since birth.** - This contrasts with Bruton XLA, which will be *just* bacterial infections (usually **since 6 months of age**). - Tx = bone marrow transplant.
Bruton X-linked agammaglobulinemia (XLA)	- X-linked recessive (as name implies) → caused by mutation in Bruton tyrosine kinase. - Isolated B cell deficiency. - Scanty/absent lymph nodes + tonsils. - Kid will have ↓ immunoglobulins of all classes (because B cells make Ab). - Q will almost always be a boy who's had only bacterial infections since 6 months of age (if they mention bacterial *and* viral, fungal, and/or protozoal, the Dx is SCID). - There is one Q on a Peds 2CK form where they say "bacterial infections since birth," but you're able to eliminate the other answers (literally 14/15 Qs will say from ~5-7 months-onward, so this is a highly annoying point to have to make, but I have to communicate it because it's on the NBME). - Tx on NBME is "monthly infusion of immunoglobulin." - Answer on USMLE will often be "deficiency of humoral immunity," rather than just "Bruton XLA."
IgA deficiency	- One of the highest yield conditions for the Step. - Most common hereditary immunodeficiency. - 9/10 Qs will give you recurrent sinopulmonary infections in someone between high school and 30s; one 2CK Peds Q has it in a kid; the Q might say the patient has a sore cheek (sinusitis) + Hx of pneumonias. - Anaphylaxis with transfusion of blood products is ultra-HY detail but often omitted from real NBME Qs because it's too easy/buzzy. - Hx of Giardia infection sometimes seen in Qs. - Atopy is really HY (asthma, seasonal allergies, eczema). - Associated with other autoimmune diseases (i.e., vitiligo, Celiac). - Celiac Ab screen falsely negative in IgA deficiency. - IgA deficiency can be associated with false-positive pregnancy tests (yes, fucking weird, but due to heterophile Ab production apparently, with zero relation to EBV). - Answer on USMLE will often be "deficiency of mucosal immunoglobulin" or "deficiency of humoral immunity," rather than just "IgA deficiency."
DiGeorge syndrome	- T cell deficiency caused by 22q11 deletion. - Failure of development of 3rd and 4th pharyngeal **pouches** (not arches) → 3rd becomes thymus + two inferior parathyroids (these structures form a triangle → 3rd); 4th becomes two superior parathyroids. - Absent thymic shadow. - Hypocalcemia due to hypoparathyroidism (positive Chvostek and Trousseau signs of hypocalcemia). - Tetralogy of Fallot (VSD, overriding aorta, pulmonic stenosis, RVH) and truncus arteriosus are HY cardiac anomalies. - Cleft lip/palate may or may not be mentioned. - Patient will have *normal* antibody levels (even though T cells activate B cells, choose normal Ab levels).

Chronic mucocutaneous candidiasis	- T cell dysfunction. Choose "deficiency of cell-mediated immunity." - Presents as recurrent candidal skin infections since birth. - Associated with other autoimmune diseases (i.e., T1DM, thyroiditis). - 17-year-old girl + recurrent candidal skin infections since birth + one year Hx of autoimmune thyroiditis + 2-yr Hx of T1DM; mechanism for infections? → answer = "deficiency of cell-mediated immunity."
Chronic granulomatous disease (CGD)	- Aka NADPH oxidase deficiency. - Usually X-linked recessive. - 1st enzyme of the respiratory burst; converts molecular O2 → superoxide. - Recurrent infections with catalase (+) organisms: *Serratia, Pseudomonas, Aspergillus, Candida, E. coli, Staph, H. pylori, Burkholderia.* - It's to my observation ~50% of Qs will simply have *S. aureus* as the organism. But the other 50% will have others from the list above. - Condition results in ↓ ability of patient to synthesize enough H2O2 to overwhelm catalase (+) organisms. - Diagnose with dihydrorhodamine test (tetrazolium blue assay is atavistic and the **wrong** answer if both are listed together, as per Step 1 NBME.)
Myeloperoxidase (MPO) deficiency	- Autosomal recessive. - Last enzyme in respiratory burst; converts H_2O_2 → hydroxyl-halide radicals (bleach). - Q will literally give you a one-liner telling you someone can't produce hydroxyl-halide radicals, and the answer is myeloperoxidase deficiency. - Q can mention myeloperoxidase deficiency in vignette, and then the answer is "impaired conversion of hydrogen peroxide to hypochlorous acid." - Usually presents as mix of *Staph aureus* and *Candida* infections. - Q on Free 120 gives mix of *S. aureus* and *Candida* infections in 6-year-old girl with AR disorder; Q wants MPO deficiency as answer, not CGD. Even though CGD can present with same organisms, the literature not only says 70% of CGD cases are XR, but if CGD occurs in children, it's usually XR, since less-frequent AR CGD is often more mild and presents later.
Wiskott-Aldrich syndrome	- X-linked recessive (*WAS* gene). - T cell cytoskeletal dysfunction. - Triad of: immunodeficiency, eczematoid skin lesions, thrombocytopenia. - ↑ serum IgA+ IgE; ↓ IgM. - 12M + *Pneumocystis* infection + Hx of nosebleeds + eczema; Q asks what cell is dysfunctional? → answer = T cell. - Answer can also be "deficiency of cell-mediated immunity."
Leukocyte adhesion deficiency (LAD)	- Deficiency of CD18/LFA-1 integrin. - ↓ ability of WBCs to leave blood vessels for sites of infection → ↓ pus. - Classically associated with delayed separation of umbilical cord. But they will only say this buzzy detail in ~50% of Qs. - The other half will say "absent pus/neutrophils in the skin." This is also buzzy, but students tend to only memorize the umbilical cord detail. - Answer can be "defective leukocyte chemotaxis" instead of just "LAD."
Chediak-Higashi syndrome	- Phagolysosomal fusion disorder with disruption of microtubule function. - Melanosome function also impaired → partial albinism / fairer skin than siblings. - Giant granules in phagocytes.
Ataxia-telangiectasia	- AR condition that results in "failure of double-stranded DNA break repair." - Cerebellar atrophy → ataxia. - Telangiectasias. - ↑ susceptibility to radiation → ↑ risk of leukemia and lymphoma.
Hyper-IgM syndrome	- Deficiency of CD40L on $CD4^+$ T cells.

	- ↓ activation of CD40 on B cells → ↓ B cell activation → ↓ ability to differentiate into plasma B cells + isotype class-switch → Ab isotype "stuck" at IgM. - Q will just give you a patient with super-elevated IgM and then the answer will simply be "deficiency of CD40 ligand."
Hyper-IgE (Job) syndrome	- Presents as **FATED** → abnormal **F**acies, staphylococcal cold **A**bscesses, retained primary **T**eeth, hyper Ig**E**, **D**ermatologic findings (e.g., eczematoid lesions).
IL-12 receptor deficiency	- Recurrent TB infections. - Important point for USMLE is that Th1 CD4+ T cells and macrophages activate each other in a cyclical loop via IL-12 and IFN-γ. - Th1 CD4+ T cell lacking IL-12 receptor cannot get activated and therefore does not secrete IFN-γ → macrophages not activated → ↓ granuloma formation.
Common variable immunodeficiency (CVID)	- Normal B cell number but defective maturation into plasma cells. - ↓ immunoglobulin production. - Will be the answer if the Q gives you an *adult* who has had recurrent sinopulmonary infections (similar to IgA deficiency), except ↓ Ig not limited to IgA; in addition, if vignette gives you atopy or Giardia, answer is IgA deficiency not CVID.
Terminal complement deficiency	- Deficiency of C5-C9 complement proteins. - Cannot form membrane attack complex (MAC). - Recurrent *Neisseria* infections (both gonococcal and meningococcal). - Q will often have deficiency of "C7" or "C8" as the answer (i.e., they choose a random terminal complement protein), or "terminal complement deficiency," or "complement-mediated immunodeficiency," or "deficiency of total hemolytic complement."

- 4F + has her MHC class I molecules investigated; which of the following processes is most likely to be seen? → answer = "degradation by proteasome"; MHC I has antigen peptides loaded in the RER before being transported to cell surface with β2-microglobulin; wrong answers are "association with invariant chain," "antigen loaded following release of invariant chain within acidified endosome," and "binding of peptides from the endocytic pathway," which refer to MHC II molecules. MHC I is present on all nucleated cells (therefore not RBCs) and presents intracellular antigens (virus) to CD8+ T cells; MHC II is present only on antigen-presenting cells (i.e., macrophages, dendritic cells, Langerhan cells, B cells) and presents exogenous antigen (bacterial products) to CD4+ T cells.
- Researcher investigating an immune structure that naturally concentrates within endosomes, lysosomes, and phagolysosomes + directly kills organisms (e.g., TB) + increasing pH within phagolysosomes decreases efficacy of the structure; Q asks, which of the following processes is primarily affected? → answer = "MHC II molecule peptide loading." If you think about it, since MHC II is expressing phagocytosed extracellular antigen, *it makes sense* that that's the MHC where endosomes play a role.

- Transgenic rats + incapable of expressing β2-microglobulin; which of the following host defenses is most likely to be altered? → answer = cytotoxic T cells; β2-microglobulin is associated with MHC I, which interacts with T cell receptor on CD8+ cytotoxic T lymphocytes.
- 4F + thymus is investigated using laser scanning microscopy; what is most likely to be seen? → answer = T lymphocyte antigen receptor selection.
- 3M + bacterial infections since 6 months of age + scanty lymph nodes/tonsils; what mechanism best explains this patient's condition? → answer = "deficiency of humoral immunity"; diagnosis is X-linked agammaglobulinemia → mutation in Bruton tyrosine kinase → decreased B cell function → decreased immunoglobulins.
- 19F + Hx of recurrent sinopulmonary infections since elementary school + asthma + Hx of Giardia infection; what mechanism best explains this patient's condition? → answer = "deficiency of humoral immunity," OR "deficiency of mucosal immunoglobulin"; diagnosis is IgA deficiency.
- 17F + Hx of cutaneous candida infections since childhood + 2-year Hx of type I diabetes mellitus + Hx of autoimmune thyroiditis; Dx? → answer = chronic mucocutaneous candidiasis → answer = "T cell dysfunction" = "impaired cell-mediated immunity" (both answers); although candida infections can be seen in diabetes, the durations don't match up here, so you know the Dx is CMC.
- 2F + numerous infections of skin + oral mucosa since birth + P/E shows numerous superficially ulcerated skin lesions + WBC 20,000 (75% neutrophils) + biopsy of skin lesion shows no neutrophils within the dermis or epidermis + culture of lesion shows *S. aureus*; Q asks, this patient's recurrent infections are most likely due to defective production of what? → answer = integrin; diagnosis is leukocyte adhesion deficiency (LAD), caused by deficient CD18/LFA-1 integrin. LFA-1 is a type of integrin; CD18 is a common subunit of integrins. LAD Qs need not mention delayed separation of umbilical cord or absence of pus (although these are two HY descriptors); sometimes the answer will be "defective leukocyte chemotaxis" instead of simply "LAD."
- 3M + severe skin infections caused by *S. aureus* since birth + no pus at infection sites + previous laboratory studies showed leukocytosis even in the absence of infection; which mechanism is impaired in this patient? → answer = "leukocyte adhesion and migration"; diagnosis is LAD.

- Researcher investigating leukocyte extravasation + chemotaxis; Q asks, which molecule is necessary for margination + rolling of leukocytes; answer = selectin; wrong answer is integrin.

Leukocyte extravasation + chemotaxis	
What the leukocyte is doing	**Important molecules required**
Margination + rolling on endothelium	L-, E-, P-selectins
Adhesion to endothelium	Integrins **(CD18/LFA-1 integrin for LAD)**
Diapedesis (extravasation)	PECAM-1
Chemotaxis to desired site	IL-8, C5a, LTB-4, kallikrein, platelet-activating factor, bacterial products

- 23F + one week of difficulty walking + blurry vision in left eye + MRI shows white matter lesions in cerebellum; which of the following immune mechanisms best describes the pathogenesis in this patient? → answer = "CD4+ T cells are activated by myelin basic protein" (MBP); diagnosis is multiple sclerosis (MS); MBP is a component of myelin. As one proposed mechanism for MS, T cells will attack MBP, via molecular mimicry, following HHV-6 infections. If you are asked to pick the cell that T cells attack in a USMLE Q for MS, answer is oligodendrocyte. For Guillain-Barre, choose Schwann cell.
- 45M + bilateral crackles + fever of 102F + *Legionella* grown from sputum culture; Q asks, which mechanism is most integral to clearing the organism from his alveoli? → answer = "T cell-mediated immunity"; *Legionella* is facultative intracellular; CD8+ cell-mediated immunity is helpful in clearing *Legionella* for this reason; very fucking HY detail for Step 1.
- 29M + splenectomy following motor vehicle accident + now has target cells; this is due to loss of what? → answer = red pulp → part of spleen (~75%) that normally clears out senescent RBCs; asplenia is rare cause of target cells for USMLE, although this shows up on one NBME Q (target cells almost always thalassemia). White pulp of the spleen (~25%) is composed of lymphoid tissue, where opsonization + phagocytosis occurs.
- 29M + splenectomy + must now receive which three vaccines? → *S. pneumo, Haemophilus influenzae type B, Neisseria meningitides* → encapsulated organisms → increased risk because normally encapsulated organisms are removed via opsonization + phagocytosis, and the spleens plays a major role in carrying out that function. C3b and IgG are two major opsonins → once bound to cell, cell now labeled for phagocytosis.
- What will be seen on blood smear post-splenectomy (or with auto-splenectomy)? → answer = Howell-Jolly bodies (RBC nuclear remnants) + target cells (although the latter usually thalassemia).

- How does the spleen relate to platelets? → splenomegaly can sequester platelets and cause thrombocytopenia; splenectomy can cause transient thrombocytosis.
- 16F + has fever and confusion + CSF culture shows gram (-) diplococci + younger brother had similar infection last year; which of the following immune disorders does this patient have? → answer = "late component of complement deficiency" → terminal complement deficiency (C6-C9) leads to recurrent *Neisseria* infections (both meningococcus and gonococcus) → defective membrane attack complex (MAC); sometimes the answer will simply just be "C7," or "C8"; sounds random, but if you know these are terminal complement then it's easy.
- 7M + three infections with *Neisseria meningitides* the past year; which of the following is most likely to be abnormal in this patient? → answer = "total hemolytic complement activation."
- 40F + receives new chemotherapeutic drug that causes myelosuppression; Q asks, in order to follow for risk of infectious complications, what should be monitored? → answer = neutrophil counts → agranulocytosis (neutropenia) is often seen with chemotherapy; also seen with various drugs such as the thionamides (propylthiocuracil + methimazole, for hyperthyroidism), ganciclovir (for CMV), methotrexate (for RA), clozapine (for schizophrenia), ticlopidine (anti-platelet); agranulocytosis will frequently present on USMLE as **mouth ulcers (mucositis),** sore throat, and fever; or they can just tell you the patient has a WBC count of, e.g., 3000/uL (NR 4-11,000) following the administration of a neutropenia-inducing agent, so you can infer the drug has dropped the WBCs.
- 35M + red spots on shins, joint pain, and fatigue + HepC + urine protein and blood; which hypersensitivity type is responsible for the renal damage? → answer = type III; diagnosis is cryoglobulinemia (immunoglobulins / Ag-Ab complexes that precipitate below 37C), causing skin lesions/ulcers, Raynaud phenomenon, muscle pain, and glomerulonephritis.
- 4F + leukocytes fail to express CD40 ligand; the immunoglobulin that predominates in this patient's serum has which biologic property? → answer = "complement activation" → diagnosis is hyper-IgM syndrome → caused by failure of expression of CD40 ligand on T cells → therefore B cells cannot be activated for immunoglobulin isotype class-switching (i.e., cannot switch from IgM to IgD, IgG, IgE, IgA). Wrong answers are: "ability to cross the placenta" = IgG; "attachment to Fc receptors on mast cells" = IgE; "secretion across mucosal epithelial cells" = IgA; "activation of B cells" = IgD.

- 42M + inflammatory bowel disease + trial of monoclonal antibody is begun; what does it most likely target? → answer = tumor necrosis factor; anti-TNF-α agents such as infliximab, adalimumab, and etanercept can be used for IBD non-responsive to NSAIDs and steroids.
- 38F + rheumatoid arthritis + treatment with NSAIDs, prednisone, and methotrexate not effective; the most appropriate next step in pharmacology blocks which cytokine? → answer = TNF-α; sounds similar to above IBD Q, but these are distinct immuno NBME Qs (let that emphasize yieldness).
- 35F + SLE + has diffuse proliferative glomerulonephritis; what is the most likely immunological mechanism associated with this patient's disease? → answer = anti-DNA/DNA immune complex deposition in the glomeruli."
- 41M + infection of dorsolateral foot that ascends lymphatically; Q asks, what is the most likely source of drainage? → answer = popliteal lymph nodes (HY). Other HY lymph node drainage points:
    - Testes + ovaries → para-aortic.
    - Foregut, midgut, hindgut → celiac, superior mesenteric, inferior mesenteric, respectively.
    - Cervix + body of uterus → external iliac.
    - Anal canal above pectinate line + bladder → internal iliac.
    - Anal canal below pectinate line + scrotum → superficial inguinal.
    - Right arm + right side of trunk above diaphragm → right lymphatic duct.
    - Left side of body (entire) + right side of body below diaphragm → thoracic duct.
    - Right lymphatic + thoracic ducts drain into the confluence of the subclavian and internal jugular veins on that respective side.
    - "Yeah but Michael, you didn't mention x, y, and z locations." Relax. As I said, the objective here is to get you HY points, not to list off minutiae that's not likely to be tested.
- 29M + HIV + receives drug that blocks viral entry into the cell; which of the following is the molecular target of this drug? → answer = chemokine receptor → refers to CCR5 as the target of maraviroc, which is an HIV entry inhibitor. Don't confuse this with enfuvirtide, which targets Gp41 and functions as an HIV fusion inhibitor.
- An investigator is comparing the live-attenuated (Sabin) vs killed (Salk) Polio vaccines; what is a common feature of these vaccines that accounts for their efficacy? → answer = "neutralizing antibodies in the circulation." The live-attenuated is given oral; it enters cells and is expressed on

MHC I for interaction with CD8+ T cells for cell-mediated immunity; there is production of neutralizing secretory IgA antibodies and activated CD8+ T cell effector in the gut, neutralizing IgG antibodies and activated CD8+ T cell effectors in the circulation, and CD8+ memory T cells; in contrast, killed vaccine does not cause cell-mediated immunity so will only cause neutralizing IgG antibodies in the circulation.

- 20F + healthy + painful swelling on the chest wall inferior to right breast + physician identifies it as a nipple; biopsy of the specimen is most likely to show what? → answer = “epithelial cells” (i.e., normal finding) → supernumerary nipples are common in women; yes, this is on the NBME for Step 1.
- 33F + shortness of breath + fever + sputum shows gram (+) diplococci; which of the following enzymes is most likely to initiate intracellular killing of the organism? → answer = “NADPH oxidase.”
- 5-month-old girl + recurrent sinopulmonary infections since birth + hypopigmentation of skin, eyes, and hair + peripheral smear shows giant granules in neutrophils and eosinophils; what’s the most likely diagnosis? → answer = Chediak-Higashi syndrome; phagolysosomal fusion disorder with disruption of microtubule function; melanosome function also impaired.
- 4F + immunodeficiency + fairer skin than siblings + giant granules seen in phagocytes; Dx? → answer = Chediak-Higashi syndrome; phagolysosomal fusion disorder with disruption of microtubule function; melanosome function also impaired.
- 2M + immunodeficiency + delayed separation of umbilical cord; Dx? → answer = leukocyte adhesion deficiency → deficiency of LFA-1/CD18 integrin.
- 2M + viral, fungal, bacterial, protozoal infections since birth + absent thymic shadow + scanty/absent lymph nodes/tonsils; Dx? → answer = severe-combined immunodeficiency (SCID) → XR type is most common and due to common gamma-chain mutation or IL-2 receptor deficiency; AR is due to adenosine deaminase (ADA) deficiency; Tx is bone marrow transplant; absent thymic shadow means T cell deficiency; scanty/absent lymphoid tissue means B cell deficiency.
- 2M + bacterial infections from ~6 months of life + scanty/absent lymph nodes/tonsils; Dx? → answer = “deficiency of humoral immunity” (Bruton X-linked agammaglobulinemia); it must be pointed out that a Q on one of the newer Peds forms has Bruton as the answer for a kid who has bacterial infections *since birth* (which throws a dagger in the classic contrast with SCID); therefore the stronger

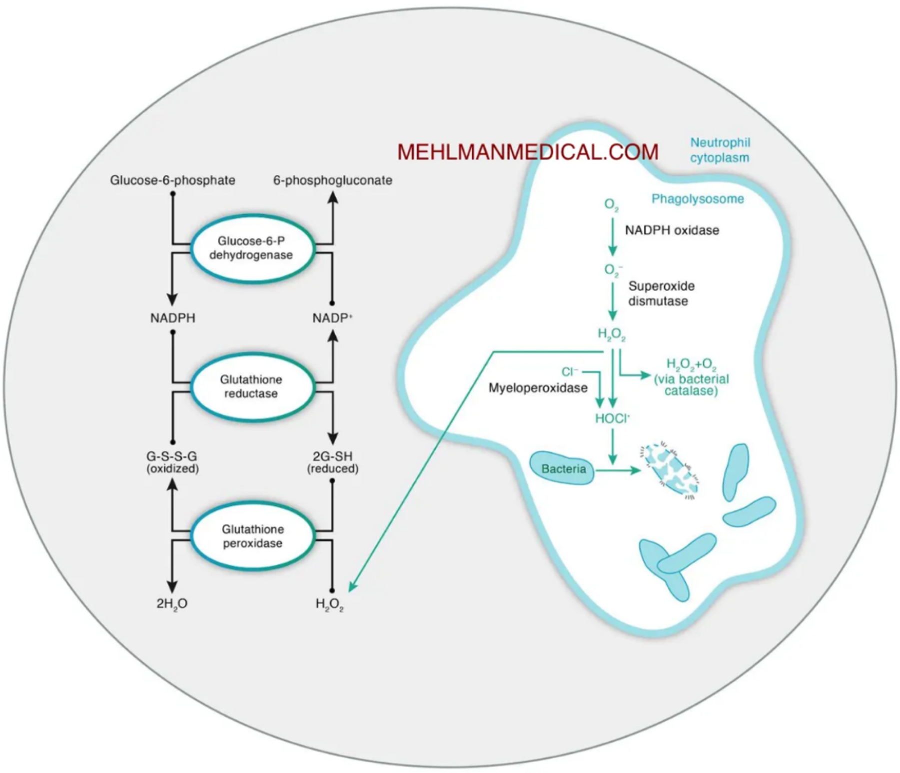
emphasis must be that Bruton is bacterial only, whereas SCID is all types of infections (viral, fungal, bacterial, protozoal); Tx = "monthly infusion of immune globulin."

- 5-month-old boy + severe diarrhea and pneumonia + thymus decreased in size + *Pneumocystis jiroveci* is isolated + decreased serum immunoglobulin concentrations + karyotype normal; diagnosis? → answer = severe combined immunodeficiency (SCID); wrong answer is DiGeorge; hypogammaglobulinemia has not been reported in DiGeorge; also, DiGeorge would have 22q11 deletion identified on karyotype analysis.
- 3M + sore throat + grey membrane at back of throat; immunization with which of the following would have prevented this? → answer = toxoid vaccine; *Corynebacterium diphtheriae* is a toxoid vaccine; presents classically with grey pseudomembrane in oropharynx and myocarditis, although vignette need not mention these.
- 34F + septic shock caused by gram (-) rod; which of the following is responsible for her low BP? → answer = "lipopolysaccharide stimulating production of tumor necrosis factor" → lipid A of lipopolysaccharide (LPS) will stimulate macrophages to release cytokines → IL-1 causes fever; TNF-alpha causes vascular permeability and low BP. Important to note that lipid A of LPS in gram (-) sepsis is an endotoxin that binds to **toll-like receptor 4 (CD14 co-receptor) on macrophages; this is in contrast to TSST toxin in toxic shock syndrome that is a superantigen that bridges MHC II and TCR.**
- 16M + nose bleeding in car accident + has nasal packing + now gets low BP; what are the immunological receptor(s) bound? → answer = MHC II and TCR → toxic shock syndrome from the cotton packing; CD14 (toll-like receptor 4 is co-receptor) is wrong answer, since that is endotoxic shock from gram (-) sepsis.
- 2M + mental retardation + fairer skin than siblings + deficiency of hydroxylase enzyme; how could this patient's presentation have been prevented? → answer = "routine newborn screening" via heel-prick test at birth for phenylketonuria (PKU; autosomal recessive; deficiency of phenylalanine hydroxylase) → tyrosine becomes essential; phenylalanine must be avoided in diet; malignant PKU = tetrahydrobiopterin (THB) deficiency (cofactor for phenylalanine hydroxylase).
- 8-month-old girl + oropharyngeal candidiasis + Hx of parainfluenza infection + 3/6 harsh systolic murmur at left sternal border; what does she have a deficiency of? → answer = T lymphocytes → possibly DiGeorge syndrome in this case (tetralogy of Fallot with murmur being a mix of VSD and

pulmonary stenosis); T lymphocyte deficiency will cause viral, fungal and protozoal infections; if the USMLE wants B cell deficiency as part of answer, they'll mention bacterial infections.

- 8-day-old newborn + truncus arteriosus + serum calcium of 7mg/dL + no thymic shadow; Dx? → answer = hypoparathyroidism (DiGeorge; 3rd and 4th **pouch** agenesis → 3rd = thymus and inferior parathyroids; 4th = superior parathyroids).
- An 8-month-old boy has recurrent respiratory tract infections since birth + hypocalcemia + broad nasal bridge; deficiency of which of the following is responsible for this patient's condition? → answer = T cell; diagnosis is DiGeorge → 22q11 deletion.
- 16F + 3-day Hx of pain and pressure over left cheek + Hx of strep pneumonias at age 6 and 10 + seasonal allergies; Dx? → answer = "impaired humoral immunity" or "deficiency of mucosal immunoglobulin"; Dx is IgA deficiency (one of the highest yield presentations for not just the Peds shelf but also USMLE); sore cheek = sinusitis; presents as recurrent sinopulmonary infections; also associated with **Hx of Giardia infection, autoimmune diseases (e.g., vitiligo), and atopy** (dry cough in winter [cough-variant asthma], hay fever in spring, eczema in summer); anaphylaxis with blood transfusion is "too easy" for most 2CK IgA deficiency Qs but will rarely show up, yes.
- 3M + recurrent OM + recently underwent placement of tympanostomy tube + mom HIV negative; Dx? → answer = "antibody deficiency" → IgA deficiency.
- 8F + fever + purpuric lesions over trunk and extremities + brother died of fulminant meningococcemia four years ago; Dx? → answer "complement system immunodeficiency" → terminal complement deficiency (C5-9) is associated with recurrent Neisseria infections (gonococcal and meningococcal).
- 18M + African American + *Neisseria meningitides* infection + had similar infection two years ago + vignette gives no other info; Q asks, this patient should be examined for which of the following conditions? → answer = "complement deficiency"; wrong answer = sickle cell disease; no reason to suspect asplenia from sickle cell.
- "What do I need to know about the respiratory burst and chronic granulomatous disease (CGD; aka NADPH oxidase deficiency)?"
    - Respiratory burst is a process by which phagocytes such as neutrophils and macrophages can produce hydrogen peroxidase and hydroxy-halide radicals (bleach) to kill microbes.

- CGD is caused by NADPH oxidase deficiency → decreased H_2O_2 production → **increased susceptibility to catalase-positive organisms**.
- Catalase is the enzyme that breaks down H_2O_2.
- Humans **without** CGD: Production of H_2O_2 via respiratory burst is >>> catalase produced by organisms → organisms are overwhelmed + die.
- Human **with** CGD: Production of H_2O_2 via respiratory burst is < catalase produced by organisms → organisms can tolerate phagocyte environment + survive.
- Organisms that produce catalase are able to neutralize *small amounts* of H_2O_2, but not large amounts. So in a typical person who does *not* have NADPH oxidase deficiency, the amount of hydrogen peroxide he or she produces far exceeds what catalase (+) organisms can handle, so therefore there's no innate susceptibility to these organisms.
- A general mnemonic for catalase-positive organisms is **C SHAPES** or **SPACESHip:**
    - *Candida, Serratia, H. pylori, Aspergillus, Pseudomonas, E. coli, S. aureus.*
    - *Staph, Psuedomonas, Aspergillus, Candida, E. coli, Serratia, H. pylori.*



- 6M + recurrent Staph infections + diarrhea; Dx? → answer = "neutrophil oxidation burst" → chronic granulomatous disease (NADPH oxidase deficiency) → infections with catalase (+) organisms (Staph infections are signature complication); SPACES → *Staph, Psuedomonas, Aspergillus,* ***Candida, E. coli, Serratia***; *Aspergillus* most common fungal infection seen in CGD; 2CK has Q where child gets Serratia sepsis as part of CGD.
- 5F + life-threatening *Staph* infections since age 3 months + WBCs normal + serum immunoglobulins normal; which of the following will best help establish the Dx? → NBME answer = "NADPH oxidase test"; **if the Q gets specific, dihydrorhodamine test is now correct over tetrazolium blue assay;** this is on either retired NBME 15 or 16 for Step 1, where the latter is wrong; the former is correct.
- 19M + type I diabetes + recurrent yeast infections + analysis shows deficiency of myeloperoxidase; Q asks most likely cause of infection susceptibility in this patient; answer = "decreased ability to produce hydroxy-halide radicals" → Dx is myeloperoxidase deficiency; enzyme normally converts hydrogen peroxide into bleach (hydroxy-halide radicals).
- 32F + has PPD test performed + large area of induration; which cell type is predominant in this area of induration? → answer = macrophages; wrong answer = cytotoxic T lymphocytes; macrophages phagocytose the TB antigen and express it on MHC II to CD4+ T cells (humoral immunity); cytotoxic T cells are CD8+ and will interact with virus expressed on MHC I (cell-mediated immunity).
- 60M + undergoes chemotherapy + has fever of 102F + neutrophil count is low + antibiotics are administered; which of the following should be given to this patient to improve his neutrophil count? → answer = granulocyte-macrophage colony-stimulating factor (GM-CSF; molgramostim), or G-CSF (filgrastim).
- 3-week-old boy born at term + experienced two weeks of respiratory distress following birth + mom has myasthenia gravis; this patient's transient respiratory distress was due to transplacental transfer of which of the following antibody types? → answer = IgG (only one that crosses the placenta).
- 14M + fever + stiff neck + non-blanching purpura on abdomen + BP of 60/35 + IV fluids and norepinephrine have limited effect; Tx? → answer = hydrocortisone; Dx = Waterhouse-Friderichsen syndrome; hemorrhagic necrosis of adrenal cortices secondary to meningococcal septicemia; non-blanching rash in the setting of meningitis = meningococcus; hydrocortisone is the answer because

cortisol is deficient in this setting → cortisol normally needed to upregulate alpha-1 receptors on arterioles, thereby permitting NE and E to do their job; that's why NE has limited effect.

- 1-month-old male + diffuse white plaques in oral cavity that bleed when scraped + papular erythematous rash with satellite lesions over the diaper area; mechanism? → answer = "decreased T lymphocyte activity" → chronic mucocutaneous candidiasis.
- 8M + super high IgM on lab report; what's the mechanism? → answer = deficiency of CD40 ligand on T cells (can't activate CD40 on B cells to induce isotype class switching) → Dx = Hyper IgM syndrome.
- 6F + recurrent Staph abscesses + eczematoid skin lesions + abnormal dentition; Dx? → answer = hyper IgE syndrome (Job syndrome) → FATED → abnormal Facies, staphylococcal cold Abscesses, retained primary Teeth, hyper IgE, Dermatologic findings (e.g., eczematoid lesions).
- 12M + eczematoid skin lesion on forehead + nosebleeds + Hx of infections; Q asks which cell is dysfunctional → **answer = T cell;** Dx = Wiskott-Aldrich syndrome (XR) → classic triad of eczematoid skin lesions + thrombocytopenia + immunodeficiency.
- 3F + dilated superficial blood vessels on face + wobbly gait; mechanism? → answer = failure of double-stranded DNA break repair" → Dx = ataxia-telangiectasia.
- 17M + 3-wk Hx of lymphadenopathy, fever, and loose stools + hepatosplenomegaly + low RBCs and WBCs + high IgM and IgG; Dx? → answer = HIV infection; wrong answers are CVID, IgA deficiency, SCID, DiGeorge, Bruton (so you can eliminate to get there); CVID is can rarely occur as young as adolescence but Dx is with IgG and IgA >2SD **less than** the mean (IgG high in this stem).
- 47M + farmer + nodule on nose + biopsy shows invasive melanoma with features of regression; which of the following best explains the mechanism of regression? → answer = "T lymphocyte-mediated cytotoxicity" → T cells can recognize and destroy cancer cells; aldesleukin is a recombinant IL-2 used for melanoma that stimulates T cells.
- 20M + Crohn disease + treated with prednisone + recovers; the mechanism of this pharmacologic therapy is suppression of which of the following? → answer = T lymphocytes; steroids lead to suppressed T cell responses; wrong answers are antibody binding, complement activity, mast cell degranulation, neutrophil function.

- 28F + squamous cell carcinoma of the cervix + biopsy of tumor cells shows HPV-18; which of the following cell types is most important for killing these virally infected cells? → answer = T cells; CD8+ T cells kill virus-infected cells; wrong answers are dendritic cells, macrophages, and plasma cells.
- 45M + has tuberculosis + which cytokine concentration is most likely to be increased in this patient? → answer = interferon-gamma (IFN-γ); IFN-γ from CD4+ T cells stimulates macrophages in order to form granulomas.
- 25M + receives bone marrow transplant + one week later develops rash, diarrhea, and increased LFTs; what's the most likely mechanism for this patient's condition? → answer = "Donor T cells reacting against host cells"; diagnosis is graft vs host disease → usually seen in bone marrow transplants, but sometimes in liver transplants; donor (graft) T cells recognize the host (recipient) antigens as foreign + cause immune response.
- 45M + bone marrow transplant 7 weeks ago + now has increased LFTs; Q asks, which of the following hypersensitivity types is most responsible? → answer = cell-mediated; graft-vs-host disease is considered to be cell-mediated hypersensitivity; wrong answers are antibody-mediated, immediate, and immune complex.
- 46F + receives liver transplant + the vessels supporting the graft appear to necrose within 15 minutes of transplantation; what's the mechanism? → answer = preformed antibodies; hyperacute rejection (graft rejection on the operating table). Acute rejection on USMLE will be up to several months; chronic rejection will be >6 months; for both acute and chronic, USMLE wants T cell response.
- 52M + infiltrating parietal lobe mass identified as astrocytoma; impaired activity of which of the following is most likely responsible? → answer = p53; this is often the answer if the USMLE asks about what needs to be mutated for there to be cancer invasion; effect of p53 mutation is "defective DNA repair"; p53 normally facilitates DNA repair by halting the cell cycle to allow time for cellular machinery to restore molecular stability.
- 31M + cellulitis + goes on to develop fever, confusion, tachycardia, and low BP + WBC 18,000/uL; which two cytokines are responsible for his condition? → answer = IL-1 and TNF-α. Really HY Q for USMLE → IL-1 causes fever associated with sepsis; TNF-α causes low BP (increased vascular permeability). If the Q gives you a vignette of sepsis and forces you to choose one molecule, the answer is TNF-α. Should be noted that students memorize IL-1 as causing fever, but this only applies

to infection/sepsis; if the vignette tells you aspirin or ibuprofen is given to decrease fever, answer = decreased effect of prostaglandin, not IL-1 (sounds easy, but I've seen students get this wrong).

- 52F + breast cancer + lost 20 lbs past month + no appetite; what is responsible for this patient's loss of appetite? → answer = "effect of cytokine" → TNF-α causes cachexia.
- Researcher compares two cell lines; one cell line has an IL-2 gene transfected; the other does not; the transfected group generates a stronger immune response; what kind of cell is responsible for this difference? → answer = T lymphocyte; IL-2 stimulates T cells.
- 40M + has a wound surgically repaired; Q asks, which of the following inflammatory mediators is most likely released from aggregated platelets in this patient during wound repair? → answer = thromboxane A2 (I guess more of a heme/onc Q, but life moves on doesn't it).
- 65F + fever of 102F + muscle pain + non-productive cough + shortness of breath; which vaccine would have prevented these findings? → answer = influenza virus.
- 20M + fever of 100.8F + non-productive cough + pale conjunctivae + blood sample agglutinates while waiting for transport to laboratory; which of the following antibody isotypes most likely caused the agglutination? → answer = IgM; *Mycoplasma pneumoniae* can cause atypical pneumonia, as well as cold autoimmune hemolytic anemia (pale conjunctivae); cold AIHA is associated with IgM targeting RBCs (cold = "**Mmm** ice cream" = Ig**M**); agglutination of RBCs occurs <37C; can also be caused by CMV; warm AIHA is IgG against RBCs and is caused classically by chronic lymphocytic leukemia (CLL).
- 61M + goes hiking + has linear vesicles on his leg; this finding is most likely caused by activation of which of the following cell types? → answer = T cells → contact dermatitis (type IV hypersensitivity) caused by poison ivy/sumac; "linear vesicles" is HY for brushing up / walking past weeds; if Q asks how to avoid this condition, answer = "avoidance of contact with weeds."
- Researcher makes a cell line that produces an inactive form of caspase. What's most likely to be the result of this? → answer = continued proliferation of cells; caspases enable apoptosis.
- 50F + has flow cytometry performed showing CD3 50%; CD4 40%; CD8 10%; CD20 50%; surface kappa 48%; surface lambda 3%; what is most predictive of clonal lymphoid proliferation in this patient? → answer = "surface kappa/lambda ratio"; CD3 = T cell marker; CD20 = B cell marker (so we know the analysis is showing us 50% T cells and 50% B cells); CD4 and CD8 are T cell markers (so 80% of the patient's T cells are CD4, and 20% are CD8); surface kappa and lambda are expressed on B cells,

where the ratio can be heavily slanted in leukemia/lymphoma. The point of this Q is to 1) be able to infer the heavy ratio of kappa/lambda means the pathology is related to this, and 2) to be aware of various T and B cell markers. Should be noted CD19 is another important B cell marker, and that rituximab is a monoclonal antibody against CD20.

- 4F + bloody diarrhea caused by *Salmonella*; which of the following factors is most responsible for neutrophil recruitment to intestinal cells? → answer = IL-8; neutrophilic chemotactic molecules include IL-8, kallikrein, platelet-activating factor, C5a, LTB-4, and bacterial products.
- 72M + fever 101 + WBCs 15,000 + CXR shows patchy infiltrate; which endogenous chemoattractant primarily causes leukocytic recruitment to his site of acute inflammation? → answer = C5a. Don't get pulled into weird distractors like filamin or *N*-formyl methionine terminal amino acid peptides.
- 2-month-old girl + feeding and developing well + pregnancy uncomplicated + temperature of 99.8F + hematocrit, platelets, and WBC count normal, but neutrophils 5%; diagnosis? → answer = congenital neutropenia"; neutrophils should be 50-55%; asked on the NBME.
- 22F + vesicular lesions on lower lip + recurrences most likely caused by viral latency where? → answer = sensory neurons.
- 4F + fever 102F + crackles and rhonchi heard in the lungs + CXR shows dextrocardia and right-sided gastric bubble; deficiency of which of the following is responsible for this patient's disorder? → answer = dynein; Dx is primary ciliary dyskinesia (Kartagener syndrome); patients present with situs inversus, sinopulmonary infections, and immotile sperm / ectopic pregnancy. Sometimes the answer can be "dynein arm" or "cilium."
- 4M + lymphadenopathy + absent granulomata seen on lymph node biopsy + acid-fast bacilli identified as *Mycobacterium avium intracellulare* + normal serum immunoglobulin concentrations and B and T cells; the patient most likely has defective expression of which of the following? → answer = interferon-gamma (IFN-γ) receptor → IFN-γ needed for granulomas.
- 60F + fever + night sweats + PPD test positive; sputum culture is most likely to show organisms inside which of the following cells? → answer = macrophages; T cells are wrong answer because they help facilitate granulomas, but the TB are phagocytosed by alveolar macrophages.
- 28F + AIDS + develops tuberculosis + which cell types is most likely to have deficient function in the tuberculous areas of lung? → answer = macrophages; CD4+ T cells normally activate macrophages via

IFN-gamma, and macrophages stimulate CD4+ T cells via IL-12; wrong answers = eosinophils, neutrophils, fibroblasts, Langerhan cells.

- 29M + PPD test positive + IFN-γ administered improves his condition; diagnosis? → answer = IL-12 receptor deficiency → causes increased susceptibility to TB infections; macrophages and CD4+ T cells have an IL-12-IFN-γ stimulatory loop → macrophages secrete IL-12, which stimulates CD4+ T cells to make IFN-γ, which in turn stimulates the alveolar macrophages to form granulomas to contain TB.
- 2-month-old girl + given vaccine that converts T cell-independent antigens into T cell-dependent antigens; which vaccine was given? → answer = *Haemophilus influenzae* type B (HiB). Only proteins can be expressed on MHC molecules in order to induce T cell response; HiB has polysaccharide capsule that will alone will not get expressed on MHC, so it is conjugated to a protein (toxoid vaccine), which in turn can be expressed on MHC, generating a T cell response.
- 2M + scaly, seborrheic eruption over scalp, palms, and diaper region + hepatosplenomegaly + generalized lymphadenopathy + x-ray of skull shows osteolytic legions + biopsy of skin lesion shows a tennis racquet-shaped bilamellar granule in cytoplasm + immunohistochemistry shows cells are CD1a positive; the abnormal cells in this patient are most likely derived from which of the following? → answer = dendritic cells; Dx = Langerhan cell histiocytosis; tennis racquet-shaped granules are called Birbeck granules. Langerhan cells are a type of dendritic cell found in the skin. Dendritic cells are antigen-presenting cells found throughout the body that, once exposed to antigen, travel to lymph nodes for interaction with B and T cells.
- 34M + chronic HepC; what mechanism best describes any hepatic damage in this patient? → answer = "Cytotoxic T cell response"; wrong answer is "direct viral cytopathic effects." Hepatitis viruses (except HepD) primarily cause hepatic damage by inducing a T cell-mediated immune response to hepatocytes; direct viral cytopathic effect is seen more with hepatitis D.
- 26M + lower back pain worse in the morning + gets better throughout the day + eczematoid lesion on his forehead; what is most likely to be seen in this patient? → answer = HLA-B27 positivity; HLA-B27 → **PAIR** → **P**soriasis, **A**nkylosing spondylitis, **I**BD, **R**eactive arthritis; never choose HLA-B27 for diagnostic purposes; USMLE just wants you to know the immunologic link for these PAIR conditions.
- 14M + abdominal mass just left to the umbilicus + chylothorax + biopsy of mass shows "starry sky appearance"; Q shows an arrow pointing to a macrophage and asks the process occurring here;

answer = apoptosis; Burkitt lymphoma can be of the jaw or intra-abdominal; “starry sky appearance” refers to a background of lymphocytes with interspersed macrophages; the latter are called “tingible body macrophages,” (not tangible) which are macrophages within germinal centers containing large amounts of apoptotic debris. Sounds outrageous and pedantic but it’s on the NBME for Step 1.

- 47M + cystic fibrosis + receives bilateral lung transplant + on immunosuppressive agents for several months + now has higher fasting glucose levels; Q asks why; answer = adverse effect of medication (tacrolimus) → HY drug that causes diabetes; tacrolimus binds to to FK506 → decreases intracellular calcineurin → decreases IL-2 transcription (cyclosporin also decreases IL-2 transcription; in contrast, sirolimus decreases *responsiveness* to IL-2).
- 42M + receives intramuscular antibiotic + three days later has rash in his arm where the drug was administered; Q asks, what kind of hypersensitivity is this? → answer = type III → diagnosis is Arthus reaction → localized skin reaction due to immune complex deposition; can be distinguished from type I hypersensitivities because type III require 3-5 days after exposure to be seen. Arthus reaction is a localized form of serum sickness; the latter is also a type III hypersensitivity due to drugs, but rather than localized to the skin, serum sickness often presents as arthritis and erythema nodosum (redness of the shins due to panniculitis [inflammation of subcutaneous fat]).
- 16F + starts taking TMP/SMX for UTI + three days later has polyarthritis; Q asks, what kind of hypersensitivity is this? → answer = type III; diagnosis is serum sickness due to sulfa drug. Serum sickness can also occur as result of infections, notably HepB/C, Rubella, Yersinia, and Chlamydia (reactive arthritis).
- 17M + leukemia + neoplastic cells do not express CD4 and CD8, however they do express T lymphocyte receptor β-chain D and J segments; Q asks, which of the following best refers to these malignant cells? → answer = “T cell thymocytes localized to thymic cortex.” Recall that T cell precursors first migrate into the thymic cortex as CD4 *and* CD8 **negative,** before differentiating into CD4 *and* CD8 positive cells. Pedantic, but asked on NBME.
- “I always get confused about the immunosuppressants. Can you consolidate some of what I need to know?”

Cyclosporine vs tacrolimus vs sirolimus			
MEHLMANMEDICAL	**Cyclosporine**	**Tacrolimus**	**Sirolimus**
Binding site	Cyclophilin	FK506	mTOR
Effect on IL-2?	↓ transcription	↓ transcription	↓ responsiveness
↓ intracellular calcineurin?	Yes	Yes	No
Causes diabetes?	No	Yes	Yes
Nephrotoxic?	Yes	Yes	No
Neurotoxic?	Yes	Yes	No
Other HY adverse effects	Gingival hyperplasia, HTN, hypertrichosis	**Diabetes highest yield**	Dyslipidemia

- "What about monoclonal antibodies? What are some worthy of memorizing for USMLE?"

Highest yield monoclonal antibodies for USMLE (i.e., sans the superfluousness)			
Drug name(s)	**Molecular target**	**Therapeutic utility**	**HY points**
Infliximab, Adalimumab, Etanercept*	Soluble TNF-α	Severe rheumatoid arthritis; IBD	*Etanercept is the odd one out → it is a recombinant TNF-α receptor that mops up TNF-α, whereas the others are monoclonal antibodies against TNF-α
Rituximab	CD20 on B cells	Lymphoma	↑ risk of progressive multifocal leukoencephalopathy (PML)
Trastuzumab	HER2/neu	HER2/neu (+) breast cancer	Cardiotoxic
Abciximab	Platelet GpIIb/IIIa	Percutaneous coronary intervention (PCI)	Students often confuse with adalimumab
Omalizumab	IgE	Severe asthma with ↑ IgE	
Denosumab	RANKL	Advanced osteoporosis	
Palivizumab	RSV F protein	RSV bronchiolitis in very rare / limited cases	Prevents RSV syncytia formation; always wrong answer for Tx on USMLE (i.e., always choose "supportive care" for RSV Tx, but just be aware the drug exists + its MOA)
Natalizumab	α4-integrin	Multiple sclerosis	↑ risk of PML
Tocilizumab	IL-6 receptor	Severe rheumatoid arthritis	
Basilixumab	IL-2 receptor	Transplant rejection	Daclizumab is another monoclonal Ab against IL-2 receptor that was once used for MS but has been discontinued
Eculizumab	Complement protein C5	Paroxysmal nocturnal hemoglobinuria (PNH)	Lowest yield one in this chart (i.e., "LY of the HY")

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